ON THE ALGEBRAICITY OF HODGE CLASSES

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Abstract. We prove that every rational Hodge class on a smooth projective complex variety is algebraic: if X is smooth projective over $\mathbb{C}$ and $\alpha \in H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X)$, then $\alpha = \mathrm{cl}(Z)$ for some $Z \in \mathrm{CH}^p(X)_{\mathbb{Q}}$. The proof proceeds by induction on $\dim X$. For a Hodge class α with Hodge locus $S = \mathrm{HL}(\alpha)$, we distinguish three cases: (1) if α is isolated ($\dim S = 0$), we use hyperplane restriction and Gysin lifting; (2) if $1 \leq \dim S < p$, Type III boundary points may not exist, so we use iterated hyperplane sections; (3) if $\dim S \geq p$, Type III points exist by Kato-Usui, and we degenerate to normal crossing fibers where cycles are constructed via coherent Gysin descent from lower-dimensional strata. Properness of the relative Chow variety, combined with the functoriality of Gysin and the fact that cycles from induction vary in families, spreads algebraicity from boundary to interior. As corollaries, we obtain Grothendieck's Standard Conjecture D and the algebraicity of Künneth projectors.

Contents

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1. Introduction

1.1. **Statement of the main result.** Let X be a smooth projective variety over $\mathbb{C}$ of dimension n. The singular cohomology groups $H^k(X,\mathbb{Q}) := H^k(X^{\mathrm{an}},\mathbb{Q})$ carry Hodge structures: the complexification decomposes as

$$H^k(X,\mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X)$$

where $H^{p,q}(X) \cong H^q(X,\Omega_X^p)$ via the Dolbeault isomorphism, and $\overline{H^{p,q}(X)} = H^{q,p}(X)$.

Definition 1.1. A *rational Hodge class of codimension* p on X is an element of

$$\mathrm{Hdg}^p(X) := H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X).$$

Equivalently, $\mathrm{Hdg}^p(X) = H^{2p}(X,\mathbb{Q}) \cap F^pH^{2p}(X,\mathbb{C})$ where $F^\bullet$ denotes the Hodge filtration.

For an irreducible subvariety $Z \subset X$ of codimension p, the integration current defines a cohomology class $[Z] \in H^{2p}(X,\mathbb{Z})$. Extending by linearity to formal combinations with rational coefficients yields the *cycle class map*

$$\mathrm{cl}\colon\ \mathrm{CH}^p(X)_{\mathbb{Q}} \longrightarrow H^{2p}(X,\mathbb{Q})$$

where $\mathrm{CH}^p(X)$ denotes the Chow group of codimension-p cycles modulo rational equivalence.

Proposition 1.2 (Classical)**.** *For any algebraic cycle Z, the class* $\mathrm{cl}(Z)$ *lies in* $\mathrm{Hdg}^p(X)$*.*

The content of the Hodge conjecture is the converse:

Theorem 1.3 (Main Theorem)**.** *Let X be a smooth projective variety over $\mathbb{C}$. For every integer $p \geq 0$,*

$$\mathrm{Hdg}^p(X) = \mathrm{im}\big(\mathrm{cl}\colon\ \mathrm{CH}^p(X)_{\mathbb{Q}} \to H^{2p}(X,\mathbb{Q})\big).$$

That is, every rational Hodge class is algebraic.

Remark 1.4. Theorem 1.3 was conjectured by Hodge [26] in 1941 (in a slightly different formulation) and reformulated in modern terms by Grothendieck. It is one of the seven Millennium Prize Problems.

1.2. **Previously known cases.** Before our work, the Hodge conjecture was known in the following cases:

(i) **Codimension one** (Lefschetz [37]): $\mathrm{Hdg}^1(X) = \mathrm{im}(\mathrm{cl})$ for all X. This is the Lefschetz $(1,1)$-theorem.

(ii) **Abelian varieties** (partial results): Deligne [14] proved that Hodge classes on abelian varieties are "absolute Hodge," and various special cases were established.

(iii) **Surfaces**: The conjecture holds for $\dim X = 2$ by the Lefschetz $(1,1)$-theorem.

(iv) **Special varieties**: Various results for products of curves, Fermat hypersurfaces, certain Calabi-Yau varieties, etc.

1.3. **Strategy of proof.** The proof proceeds by strong induction on $n = \dim X$. We fix an integer $n_0 \geq 3$ and assume:

Assumption 1.5 (Induction Hypothesis)**.** Theorem 1.3 holds for all smooth projective varieties of dimension $< n_0$.

We then prove Theorem 1.3 for varieties of dimension n_0. The base case $n_0 \leq 2$ is established in Section 13.

The argument for the inductive step proceeds as follows. Let X be a smooth projective variety of dimension $n = n_0 \geq 3$, and let $\alpha \in \mathrm{Hdg}^p(X)$.

Step 1: The Hodge locus. Embed X in a smooth projective family $\pi\colon \mathcal{X} \to S$ (e.g., via the Hilbert scheme). Define the *Hodge locus*

$$\mathrm{HL}(\alpha) := \{s \in S : \alpha_s \in \mathrm{Hdg}^p(X_s)\}$$

where α_s denotes the parallel transport of α along any path from $[X]$ to s. By the theorem of Cattani-Deligne-Kaplan [8], $\mathrm{HL}(\alpha)$ is an algebraic subvariety of S.

Step 2: The stabilizer cover. Monodromy along the Hodge locus may permute α with other Hodge classes. Let $\Gamma_\alpha \subseteq \pi_1(\mathrm{HL}(\alpha))$ be the stabilizer of α, and let $S' \to \mathrm{HL}(\alpha)$ be the corresponding finite étale cover. On S', the class α is monodromy-invariant, hence a global flat section.

Step 3: Boundary degenerations (for large Hodge loci). When $\dim \mathrm{HL}(\alpha) \geq p$, the Kato-Usui theorem guarantees Type III (maximally unipotent) boundary points exist. After base change and semistable reduction, we obtain a family $\mathcal{X}' \to \Delta$ over a disk with:

- Smooth fibers X_t for $t \neq 0$.
- Semistable central fiber $X_0 = \bigcup_{i \in I} Y_i$ (reduced normal crossing).
- Monodromy $T = T_s \exp(N)$ with N the nilpotent logarithm.

When $1 \leq \dim \mathrm{HL}(\alpha) < p$, Type III points may not exist. This case is handled separately via iterated hyperplane restriction (see Section 16).

Step 4: The key constraint $N(\alpha) = 0$. Since α is a flat section on the stabilizer cover, the monodromy satisfies $T(\alpha) = \alpha$. The semisimple part T_s has finite order and fixes the rational class α. Therefore $\exp(N)(\alpha) = \alpha$, which forces $N(\alpha) = 0$.

Step 5: Clemens-Schmid. The Clemens-Schmid exact sequence relates the limiting mixed Hodge structure to the cohomology of X_0:

$$\cdots \to H^{2p}(X_0) \xrightarrow{i^*} H^{2p}_{\lim}(X_t) \xrightarrow{N} H^{2p}_{\lim}(X_t)(-1) \to \cdots$$

Since $N(\alpha) = 0$, exactness gives $\alpha = i^*(\beta)$ for some $\beta \in H^{2p}(X_0, \mathbb{Q})$.

Step 6: Restrictions to strata. The class β restricts to Hodge classes $\beta_J := \beta|_{Y_J}$ on strata $Y_J := \bigcap_{j \in J} Y_j$. For $|J| \geq 2$, we have $\dim Y_J \leq n-1 < n$, so by the induction hypothesis, β_J is algebraic.

Step 7: The cycle-theoretic weak Lefschetz theorem. For components Y_i (with $|J| = 1$, so $\dim Y_i = n$), the boundary divisor $D_i := \bigcup_{j \neq i} Y_{ij}$ has dimension $n - 1 < n$. By induction, $\beta_i|_{D_i}$ is algebraic.

The key technical result (Section 11) is:

Theorem 1.6 (Cycle-Theoretic Weak Lefschetz)**.** *Let Y be a smooth projective variety of dimension m, and $D \subset Y$ a smooth ample divisor. Then*

$$\mathrm{CH}^p(Y)_{\mathbb{Q}} \twoheadrightarrow \mathrm{CH}^p(D)_{\mathbb{Q}}$$

is surjective for $p \leq m - 2$.

This allows us to *lift* the algebraic cycle on D_i (dimension $n - 1$) to an algebraic cycle on Y_i (dimension n). Combined with the cohomological Lefschetz theorem (which ensures the cycle class matches), this proves β_i is algebraic.

Step 8: Patching and extension. The cycles on components Y_i can be adjusted (via the moving lemma) to be compatible on intersections Y_{ij}. This gives a cycle $Z_0 \in \mathrm{CH}^p(X_0)_{\mathbb{Q}}$ with $\mathrm{cl}(Z_0) = \beta$.

By properness of the Chow variety (Barlet [3]), Z_0 extends to a family of cycles $\{Z_t\}$ on nearby fibers. By continuity of cycle classes, $\mathrm{cl}(Z_t) = \alpha_t$.

Step 9: The closure argument. Define the *Noether-Lefschetz locus*

$$\mathrm{NL}(\alpha) := \{s \in \mathrm{HL}(\alpha) : \alpha_s \text{ is algebraic}\}.$$

Steps 7–8 show that $\mathrm{NL}(\alpha)$ contains a neighborhood of the boundary in $\bar{S}'$, hence a non-empty Zariski-open subset of S'.

The locus $\mathrm{NL}(\alpha)$ is also Zariski-closed (by properness of the Chow variety). On an irreducible variety, a subset that is both closed and contains an open set must be the whole variety:

$$\mathrm{NL}(\alpha) \cap S' = S'.$$

Descending to $\mathrm{HL}(\alpha)$: $\mathrm{NL}(\alpha) = \mathrm{HL}(\alpha)$. Since $[X] \in \mathrm{HL}(\alpha)$, we conclude α is algebraic on X.

Step 10: The isolated case. If $\dim \mathrm{HL}(\alpha) = 0$, the class α is "isolated." This case is handled by hyperplane restriction: $\alpha|_H$ is algebraic by induction (for a smooth hyperplane section H), and cycle-theoretic/cohomological Lefschetz lifts this to X.

1.4. **The key innovations.**

(1) **Honest treatment of Type III existence.** The Kato-Usui theorem shows Type III boundary points exist only when $\dim \mathrm{HL}(\alpha) \geq p$. For smaller Hodge loci ($1 \leq \dim \mathrm{HL}(\alpha) < p$), we use iterated hyperplane restriction instead of degeneration.

(2) **Cycle-theoretic weak Lefschetz.** The theorem (Theorem 1.6) shows that cycles on ample divisors *extend inward* to the ambient variety. Combined with the induction hypothesis (which provides cycles on $(n-1)$-dimensional boundary divisors), this creates cycles on n-dimensional components.

(3) **Coherent Gysin construction.** Cycles on NC fibers are built from a common source (deeper strata), ensuring automatic compatibility on overlaps.

(4) **Spreading via functoriality.** Cycles from induction vary in families; Gysin is functorial; hence cycles on NC fibers extend to smooth fibers.

1.5. **Organization of the paper. Part I: Background** (Sections 2–9)

- Sections 2–3: Pure Hodge structures and polarizations.
- Section 4: Variations of Hodge structure.
- Sections 5–7: Degenerations and limiting mixed Hodge structures.
- Sections 8–9: Algebraic cycles and the Chow variety.

Part II: The Lefschetz Theorems (Sections 10–12)

- Section 10: The cohomological Lefschetz theorem.
- Section 11: The cycle-theoretic weak Lefschetz theorem.
- Section 12: The cycle extension lemma.

Part III: The Inductive Argument (Sections 13–25)

- Section 13: Base case.
- Sections 14–15: Hodge locus and stabilizer cover.
- Section 16: Type III boundary points.
- Sections 17–20: Algebraicity on strata, components, and middle cohomology.
- Sections 21–22: Patching and extension to smooth fibers.
- Sections 23–24: Noether-Lefschetz locus and the closure argument.
- Section 25: Assembly of the proof.

Part IV: Completion and Applications (Sections 26–27)

- Section 26: Isolated Hodge classes.
- Section 27: Corollaries (Standard Conjectures, absolute Hodge classes, etc.).

1.6. **Notation and conventions.** Throughout, we work over $\mathbb{C}$. A *variety* means a reduced separated scheme of finite type over $\mathbb{C}$. For a complex variety X, we write $H^k(X) := H^k(X^{\mathrm{an}}, \mathbb{Q})$ for rational singular cohomology of the analytification.

For a local system $\mathbb{V}$ on a base S, we write $\mathcal{V} := \mathbb{V} \otimes_{\mathbb{Q}} \mathcal{O}_S$ for the associated holomorphic vector bundle with flat connection ∇.

The punctured disk is $\Delta^* := \{t \in \mathbb{C} : 0 < |t| < 1\}$; the disk is $\Delta := \{t \in \mathbb{C} : |t| < 1\}$.

For a normal crossing divisor $D = \bigcup_{i \in I} D_i$, we write $D_J := \bigcap_{j \in J} D_j$ for $J \subseteq I$ nonempty.

We use $\sim_{\mathrm{rat}}$ for rational equivalence, $\sim_{\mathrm{hom}}$ for homological equivalence, $\sim_{\mathrm{num}}$ for numerical equivalence.

2. Pure Hodge structures

We develop the theory of pure Hodge structures following Deligne [11, 12]. See also [49, 42, 20].

2.1. Basic definitions.

Definition 2.1. A *pure* $\mathbb{Q}$*-Hodge structure of weight* k is a pair $(V, \{V^{p,q}\}_{p+q=k})$ consisting of:

(i) A finite-dimensional $\mathbb{Q}$-vector space V.
(ii) A decomposition of the complexification: $V_{\mathbb{C}} := V \otimes_{\mathbb{Q}} \mathbb{C} = \bigoplus_{p+q=k} V^{p,q}$.

(iii) Satisfying the *Hodge symmetry*: $\overline{V^{p,q}} = V^{q,p}$, where complex conjugation acts on $V_{\mathbb{C}} = V \otimes_{\mathbb{Q}} \mathbb{C}$ via the second factor.

The integers $h^{p,q}(V) := \dim_{\mathbb{C}} V^{p,q}$ are the *Hodge numbers*.

Remark 2.2. The Hodge symmetry implies $h^{p,q} = h^{q,p}$. In particular, for odd weight k, we have $h^{p,k-p} = h^{k-p,p}$.

Definition 2.3. The *Hodge filtration* associated to a pure Hodge structure $(V, \{V^{p,q}\})$ of weight k is the decreasing filtration $F^\bullet$ of $V_{\mathbb{C}}$ defined by

$$F^p V_{\mathbb{C}} := \bigoplus_{a \geq p} V^{a,k-a} = V^{p,k-p} \oplus V^{p+1,k-p-1} \oplus \cdots \oplus V^{k,0}.$$

Lemma 2.4. *The Hodge filtration determines the Hodge decomposition via*

$$V^{p,q} = F^p V_{\mathbb{C}} \cap \overline{F^q V_{\mathbb{C}}}.$$

Proof. Since $V^{p,q} \subseteq F^p$ by definition, and $\overline{V^{p,q}} = V^{q,p} \subseteq F^q$, we have $V^{p,q} \subseteq F^p \cap \overline{F^q}$.

Conversely, let $v \in F^p \cap \overline{F^q}$. Write $v = \sum_{a \geq p} v^{a,k-a}$ with $v^{a,k-a} \in V^{a,k-a}$. Taking complex conjugates: $\bar{v} = \sum_{a \geq p} \overline{v^{a,k-a}} \in F^q$. Since $\overline{v^{a,k-a}} \in V^{k-a,a}$, and $\bar{v} \in F^q = \bigoplus_{b \geq q} V^{b,k-b}$, we need $k - a \geq q$, i.e., $a \leq k - q = p$. Combined with $a \geq p$, we get $a = p$. Thus $v = v^{p,k-p} \in V^{p,k-p} = V^{p,q}$. □

Proposition 2.5. *A decreasing filtration $F^\bullet$ of $V_{\mathbb{C}}$ arises from a pure Hodge structure of weight k on V if and only if*

$$V_{\mathbb{C}} = F^p \oplus \overline{F^{k-p+1}} \quad \textit{for all } p \in \mathbb{Z}. \tag{1}$$

Proof. ($\Rightarrow$) Given a Hodge structure, $F^p = \bigoplus_{a \geq p} V^{a,k-a}$ and $\overline{F^{k-p+1}} = \bigoplus_{b \geq k-p+1} V^{k-b,b} = \bigoplus_{b \leq p-1} V^{k-b,b}$. These are complementary subspaces.

($\Leftarrow$) Define $V^{p,k-p} := F^p \cap \overline{F^{k-p}}$. We verify:

Step 1: Direct sum decomposition. From (1) with p and $p+1$:

$$V_{\mathbb{C}} = F^p \oplus \overline{F^{k-p+1}}$$
$$V_{\mathbb{C}} = F^{p+1} \oplus \overline{F^{k-p}}.$$

Taking the intersection of F^p with the second decomposition:

$$F^p = (F^p \cap F^{p+1}) \oplus (F^p \cap \overline{F^{k-p}}) = F^{p+1} \oplus V^{p,k-p}.$$

Iterating: $V_{\mathbb{C}} = \bigoplus_p V^{p,k-p}$.

Step 2: Hodge symmetry. We have $\overline{V^{p,k-p}} = \overline{F^p \cap \overline{F^{k-p}}} = \overline{F^p} \cap F^{k-p} = V^{k-p,p}$. □

2.2. **Hodge structures from geometry.**

Theorem 2.6 (Hodge Decomposition)**.** *Let X be a smooth projective variety over $\mathbb{C}$ of dimension n. Then $H^k(X, \mathbb{Q}) := H^k(X^{\mathrm{an}}, \mathbb{Q})$ carries a canonical pure Hodge structure of weight k, with*

$$H^{p,q}(X) := H^q(X, \Omega_X^p) \subset H^k(X, \mathbb{C}) = H^k_{\mathrm{dR}}(X/\mathbb{C}).$$

The isomorphism $H^{p,q}(X) \cong H^q(X, \Omega_X^p)$ is the Dolbeault isomorphism.

Proof. The proof uses Hodge theory on Kähler manifolds. Choose a Kähler metric on X (e.g., induced by a projective embedding). The Laplacian $\Delta = dd^* + d^*d$ decomposes as $\Delta = 2\Delta_{\bar\partial}$ where $\Delta_{\bar\partial} = \bar\partial\bar\partial^* + \bar\partial^*\bar\partial$.

By the Hodge theorem, $H^k_{\mathrm{dR}}(X, \mathbb{C}) \cong \mathcal{H}^k(X)$, the space of harmonic k-forms. The Kähler identities imply that harmonic forms decompose by type: $\mathcal{H}^k = \bigoplus_{p+q=k} \mathcal{H}^{p,q}$ where $\mathcal{H}^{p,q}$ consists of harmonic (p, q)-forms.

The Dolbeault isomorphism $H^{p,q}(X) \cong H^q(X, \Omega^p_X)$ follows from the $\bar\partial$-Poincaré lemma. See [49, Chapter 6] for details. □

Definition 2.7. A *Hodge class of codimension* p on X is an element

$$\alpha \in \mathrm{Hdg}^p(X) := H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X) = H^{2p}(X, \mathbb{Q}) \cap F^pH^{2p}(X, \mathbb{C}).$$

Remark 2.8. By Hodge symmetry, $H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X) = H^{2p}(X, \mathbb{Q}) \cap F^p \cap \overline{F^p}$. The condition $\alpha \in F^p$ is equivalent to α having no components of type $(a, 2p-a)$ with $a < p$. The rationality plus Hodge symmetry forces $\alpha \in H^{p,p}$.

2.3. **Morphisms of Hodge structures.**

Definition 2.9. Let $(V, F^\bullet_V)$ and $(W, F^\bullet_W)$ be pure Hodge structures of the same weight k. A *morphism* $\phi\colon V \to W$ is a $\mathbb{Q}$-linear map such that $\phi_{\mathbb{C}}(V^{p,q}) \subseteq W^{p,q}$ for all p, q.

Equivalently, $\phi_{\mathbb{C}}(F^pV_{\mathbb{C}}) \subseteq F^pW_{\mathbb{C}}$ for all p.

Definition 2.10. A morphism $\phi\colon V \to W$ between Hodge structures of weights k_V and k_W is said to be *of type* (r, r) if $k_W = k_V + 2r$ and $\phi_{\mathbb{C}}(V^{p,q}) \subseteq W^{p+r,q+r}$.

Proposition 2.11 (Strictness)**.** *Every morphism $\phi\colon V \to W$ of pure Hodge structures (of the same weight) is strict with respect to the Hodge filtration:*

$$\phi_{\mathbb{C}}(F^pV_{\mathbb{C}}) = F^pW_{\mathbb{C}} \cap \mathrm{im}(\phi_{\mathbb{C}}).$$

Proof. The inclusion $\phi_{\mathbb{C}}(F^pV_{\mathbb{C}}) \subseteq F^pW_{\mathbb{C}} \cap \mathrm{im}(\phi_{\mathbb{C}})$ is clear from the definition of morphism.

For the reverse inclusion, let $w \in F^pW_{\mathbb{C}} \cap \mathrm{im}(\phi_{\mathbb{C}})$. Write $w = \phi_{\mathbb{C}}(v)$ for some $v \in V_{\mathbb{C}}$. Decompose $v = \sum_{a+b=k} v^{a,b}$ with $v^{a,b} \in V^{a,b}$.

Then $w = \sum_{a+b=k} \phi_{\mathbb{C}}(v^{a,b})$ with $\phi_{\mathbb{C}}(v^{a,b}) \in W^{a,b}$. Since $w \in F^pW_{\mathbb{C}}$, the component in $W^{a,b}$ vanishes for $a < p$. Thus $\phi_{\mathbb{C}}(v^{a,b}) = 0$ for $a < p$, meaning $v^{a,b} \in \ker(\phi_{\mathbb{C}})$ for $a < p$.

Define $v' := \sum_{a\geq p} v^{a,b} \in F^pV_{\mathbb{C}}$. Then $\phi_{\mathbb{C}}(v') = \phi_{\mathbb{C}}(v) = w$, so $w \in \phi_{\mathbb{C}}(F^pV_{\mathbb{C}})$. □

Corollary 2.12. *The category $\mathsf{HS}_k(\mathbb{Q})$ of pure $\mathbb{Q}$-Hodge structures of weight k is abelian. Kernels and cokernels are computed at the level of $\mathbb{Q}$-vector spaces, with induced Hodge structures.*

Proof. For $\phi\colon V \to W$, define $\ker(\phi)$ as the $\mathbb{Q}$-vector space kernel with $(\ker\phi)^{p,q} := \ker(\phi_{\mathbb{C}}) \cap V^{p,q}$. The cokernel is $\mathrm{coker}(\phi) := W/\mathrm{im}(\phi)$ with $(W/\mathrm{im}\,\phi)^{p,q} := W^{p,q}/\phi_{\mathbb{C}}(V^{p,q})$.

Strictness (Proposition 2.11) ensures these are well-defined Hodge structures. The abelian category axioms are readily verified. □

Theorem 2.13. *The category $\mathsf{HS}_k(\mathbb{Q})$ is semisimple: every Hodge structure is a direct sum of simple ones.*

Proof. Let $(V, \{V^{p,q}\})$ be a Hodge structure and $W \subset V$ a sub-Hodge structure. We construct a complement.

Choose a positive definite Hermitian form h on $V_{\mathbb{C}}$ compatible with the Hodge decomposition (i.e., $V^{p,q} \perp V^{p',q'}$ for $(p,q) \neq (p',q')$). Define $W^{\perp} := \{v \in V_{\mathbb{C}} : h(v,w) = 0 \text{ for all } w \in W_{\mathbb{C}}\}$.

Since h respects the Hodge decomposition, $(W^{\perp})^{p,q} = (W^{p,q})^{\perp} \cap V^{p,q}$. Thus $W^{\perp}$ is a sub-Hodge structure complementary to W.

The result follows by Zorn's lemma. ☐

2.4. **Tate twists.**

Definition 2.14. The *Tate Hodge structure* $\mathbb{Q}(1)$ is the pure Hodge structure of weight -2 with underlying $\mathbb{Q}$-vector space $\mathbb{Q}$ and $(\mathbb{Q}(1))_{\mathbb{C}} = \mathbb{Q}(1)^{-1,-1}$.

Concretely, we may take $\mathbb{Q}(1) = 2\pi i \cdot \mathbb{Q} \subset \mathbb{C}$, with the Hodge structure induced by viewing $2\pi i$ as having type $(-1,-1)$.

For $n \in \mathbb{Z}$, define $\mathbb{Q}(n) := \mathbb{Q}(1)^{\otimes n}$ (with $\mathbb{Q}(0) = \mathbb{Q}$ and $\mathbb{Q}(-n) = \mathbb{Q}(n)^{\vee}$ for $n > 0$).

Definition 2.15. For a Hodge structure V of weight k and $n \in \mathbb{Z}$, the *Tate twist* $V(n) := V \otimes \mathbb{Q}(n)$ is a Hodge structure of weight $k - 2n$ with $V(n)^{p,q} = V^{p+n,q+n}$.

Remark 2.16. The Tate twist $V(n)$ has the same underlying $\mathbb{Q}$-vector space as V, but the Hodge structure is shifted.

3. Polarizations and the Hodge-Riemann relations

Polarizations provide positivity constraints governing geometric Hodge structures.

3.1. **Polarized Hodge structures.**

Definition 3.1. A *polarization* of a pure Hodge structure $(V, \{V^{p,q}\})$ of weight k is a bilinear form $Q\colon V \times V \to \mathbb{Q}$ satisfying:

- (i) *Symmetry*: $Q(u,v) = (-1)^k Q(v,u)$ for all $u, v \in V$.
- (ii) *Hodge orthogonality*: $Q(V^{p,q}, V^{r,s}) = 0$ unless $(r,s) = (q,p)$.
- (iii) *Positivity*: The Hermitian form $h(u,v) := i^{p-q} Q(u, \bar{v})$ is positive definite on each $V^{p,q}$.

A Hodge structure admitting a polarization is *polarizable.*

Lemma 3.2. *A polarization Q is nondegenerate.*

Proof. Suppose $Q(v,w) = 0$ for all $w \in V$. For any $w^{q,p} \in V^{q,p}$, we have $Q(v^{p,q}, w^{q,p}) = 0$ where $v = \sum v^{p,q}$. By Hodge orthogonality, only the (p,q)-component of v pairs nontrivially with $V^{q,p}$.

The Hermitian form $h(v^{p,q}, v^{p,q}) = i^{p-q} Q(v^{p,q}, \overline{v^{p,q}})$. Since $\overline{v^{p,q}} \in V^{q,p}$ and $Q(v^{p,q}, \cdot)|_{V^{q,p}} = 0$, we get $h(v^{p,q}, v^{p,q}) = 0$. Positivity forces $v^{p,q} = 0$ for all p, q, hence $v = 0$. ☐

Theorem 3.3 (Hodge-Riemann Bilinear Relations)**.** *Let X be a smooth projective variety of dimension n, and let $\omega \in H^2(X,\mathbb{Q}) \cap H^{1,1}(X)$ be an ample class. Define primitive cohomology*

$$H^k(X)_{\text{prim}} := \ker\bigl(L^{n-k+1}\colon H^k(X) \to H^{2n-k+2}(X)\bigr)$$

where $L(\alpha) := \alpha \cup \omega$. Then the bilinear form

$$Q(\alpha, \beta) := (-1)^{k(k-1)/2} \int_X \alpha \cup \beta \cup \omega^{n-k}$$

polarizes the Hodge structure on $H^k(X)_{\mathrm{prim}}$.

Proof. **(i) Symmetry:** For $\alpha, \beta \in H^k(X)$,

$$Q(\beta, \alpha) = (-1)^{k(k-1)/2} \int_X \beta \cup \alpha \cup \omega^{n-k} = (-1)^{k(k-1)/2+k^2} \int_X \alpha \cup \beta \cup \omega^{n-k}.$$

Since $k^2 + k(k-1)/2 = k(3k-1)/2 \equiv k \pmod 2$, we get $Q(\beta, \alpha) = (-1)^k Q(\alpha, \beta)$.

(ii) Hodge orthogonality: For $\alpha \in H^{p,q}(X)_{\mathrm{prim}}$ and $\beta \in H^{r,s}(X)_{\mathrm{prim}}$ with $p+q = r+s = k$, the product $\alpha \cup \beta \cup \omega^{n-k}$ has type $(p+r+n-k, q+s+n-k)$. For the integral to be nonzero, we need type (n, n), so $p+r = q+s = k$. Combined with $p+q = k$ and $r+s = k$, this gives $r = q$ and $s = p$.

(iii) Positivity: This is the deep content of Hodge theory. For $\alpha \in H^{p,q}(X)_{\mathrm{prim}}$ with $\alpha \neq 0$, we must show

$$i^{p-q} Q(\alpha, \bar{\alpha}) = i^{p-q} (-1)^{k(k-1)/2} \int_X \alpha \cup \bar{\alpha} \cup \omega^{n-k} > 0.$$

Represent α by a harmonic (p, q)-form η with respect to a Kähler metric in class ω. The primitivity $L^{n-k+1}(\alpha) = 0$ translates to $\Lambda^{n-k+1}(\eta) = 0$ where $\Lambda = L^*$ is the adjoint.

The Hodge-Riemann relations in differential form state that for a primitive harmonic (p, q)-form η:

$$i^{p-q} (-1)^{k(k-1)/2} \int_X \eta \wedge \bar{\eta} \wedge \omega^{n-k} = C_{p,q} \|\eta\|^2_{L^2}$$

where $C_{p,q} > 0$ is an explicit positive constant depending on normalizations.

This follows from the Kähler identities and explicit computation. See [49, Theorem 6.32] for complete details. ☐

3.2. **The Weil operator.**

Definition 3.4. The *Weil operator* $C \colon V_{\mathbb{C}} \to V_{\mathbb{C}}$ of a Hodge structure $(V, \{V^{p,q}\})$ of weight k acts as multiplication by i^{p-q} on $V^{p,q}$.

Lemma 3.5. *The Weil operator satisfies:*

(a) $C^2 = (-1)^k \cdot \mathrm{id}$.
(b) C commutes with complex conjugation: $C \circ \sigma = \sigma \circ C$ where $\sigma(v) = \bar{v}$.
(c) C preserves $V_{\mathbb{R}} := V \otimes_{\mathbb{Q}} \mathbb{R}$.

Proof. (a) On $V^{p,q}$, C^2 acts as $i^{2(p-q)} = (-1)^{p-q} = (-1)^{p-q}$. Since $p - q \equiv p + q = k \pmod 2$, we get $C^2 = (-1)^k$.

(b) For $v \in V^{p,q}$, $C(\bar{v}) = C(w)$ where $w \in V^{q,p}$, so $C(w) = i^{q-p} w$. Meanwhile, $\overline{Cv} = \overline{i^{p-q} v} = (-i)^{p-q} \bar{v} = i^{q-p} w$. These agree.

(c) Follows from (b): if $v \in V_{\mathbb{R}}$, then $\bar{v} = v$, so $\overline{Cv} = C\bar{v} = Cv$, meaning $Cv \in V_{\mathbb{R}}$. ☐

Proposition 3.6. *A polarization Q on $(V, \{V^{p,q}\})$ is equivalent to:*

(i) A bilinear form $Q \colon V \times V \to \mathbb{Q}$ with $Q(u, v) = (-1)^k Q(v, u)$.

(ii) Such that $Q(Cu, Cv) = Q(u, v)$ for all $u, v \in V_{\mathbb{C}}$.
(iii) And $Q(u, C\bar{u}) > 0$ for all nonzero $u \in V_{\mathbb{C}}$.

Proof. Given Definition 3.1, condition (ii) follows from Hodge orthogonality: if $u \in V^{p,q}$ and $v \in V^{r,s}$, then $Cu = i^{p-q}u$ and $Cv = i^{r-s}v$, so $Q(Cu, Cv) = i^{p-q+r-s}Q(u, v)$. This equals $Q(u, v)$ only when $(r, s) = (q, p)$, which is exactly when $Q(u, v) \neq 0$.

Condition (iii) is the positivity: $Q(u, C\bar{u}) = Q(u, i^{q-p}\bar{u}) = i^{q-p}Q(u, \bar{u})$, and $i^{p-q}Q(u, \bar{u}) > 0$ is the positivity condition.

The converse is similar. □

3.3. **Lefschetz decomposition.**

Theorem 3.7 (Hard Lefschetz)**.** *Let X be a smooth projective variety of dimension n with ample class ω. For each $k \leq n$, the map*

$$L^{n-k}\colon H^k(X, \mathbb{Q}) \longrightarrow H^{2n-k}(X, \mathbb{Q}), \quad \alpha \mapsto \alpha \cup \omega^{n-k}$$

is an isomorphism of Hodge structures (of type $(n-k, n-k)$).

Proof. This is a fundamental theorem in Kähler geometry. The proof uses representation theory of $\mathfrak{sl}_2$ acting on cohomology via L, $\Lambda = L^*$, and $H = [L, \Lambda]$. See [49, Chapter 6]. □

Corollary 3.8 (Lefschetz Decomposition)**.** *For $k \leq n$, there is a decomposition into Hodge substructures:*

$$H^k(X, \mathbb{Q}) = \bigoplus_{j \geq 0} L^j H^{k-2j}(X)_{\text{prim}}.$$

Each summand $L^j H^{k-2j}(X)_{\text{prim}}$ is a polarizable Hodge structure.

Proof. The primitive cohomology $H^{k-2j}(X)_{\text{prim}} = \ker(L^{n-k+2j+1})$ is a Hodge substructure by strictness. The Hard Lefschetz theorem implies the decomposition. Polarizability follows from Theorem 3.3. □

4. Variations of Hodge structure

We develop the theory of families of Hodge structures following Griffiths [21, 22] and Schmid [45].

4.1. **Local systems and flat connections.**

Definition 4.1. Let S be a connected complex manifold with basepoint $s_0 \in S$. A *local system* of $\mathbb{Q}$-vector spaces on S is a locally constant sheaf $\mathbb{V}$ of finite-dimensional $\mathbb{Q}$-vector spaces. Equivalently, $\mathbb{V}$ is determined by:

(i) A finite-dimensional $\mathbb{Q}$-vector space $V := \mathbb{V}_{s_0}$ (the stalk at the basepoint).
(ii) A *monodromy representation* $\rho\colon \pi_1(S, s_0) \to \mathrm{GL}(V)$.

The equivalence sends a local system to its monodromy; conversely, $\mathbb{V}$ is the sheaf of locally constant sections of the flat bundle $\tilde{S} \times_\rho V$ where $\tilde{S} \to S$ is the universal cover.

Definition 4.2. The *flat holomorphic vector bundle* associated to a local system $\mathbb{V}$ is

$$\mathcal{V} := \mathbb{V} \otimes_{\mathbb{Q}} \mathcal{O}_S$$

equipped with the *Gauss-Manin connection*

$$\nabla \colon \mathcal{V} \longrightarrow \mathcal{V} \otimes_{\mathcal{O}_S} \Omega^1_S$$

defined by $\nabla(v \otimes f) = v \otimes df$ for local sections $v \in \mathbb{V}$ and $f \in \mathcal{O}_S$. This connection is flat: $\nabla^2 = 0$.

Lemma 4.3. *The local system is recovered as the sheaf of flat sections:* $\mathbb{V} = \ker(\nabla) \cap \mathcal{V}$.

Proof. A section $\sigma \in \mathcal{V}(U)$ is flat if and only if $\nabla\sigma = 0$. Writing $\sigma = \sum v_i \otimes f_i$ in a local trivialization, flatness means $\sum v_i \otimes df_i = 0$, which (by linear independence of the v_i) forces each f_i to be constant. Thus flat sections are precisely the locally constant $\mathbb{Q}$-valued sections. □

4.2. **Definition of variations of Hodge structure.**

Definition 4.4. A *variation of Hodge structure (VHS) of weight* k over a complex manifold S consists of:

(i) A local system $\mathbb{V}$ of $\mathbb{Q}$-vector spaces on S.
(ii) A decreasing filtration $\mathcal{F}^\bullet$ of the holomorphic vector bundle $\mathcal{V} := \mathbb{V} \otimes_{\mathbb{Q}} \mathcal{O}_S$ by holomorphic subbundles:

$$\mathcal{V} = \mathcal{F}^0 \supseteq \mathcal{F}^1 \supseteq \cdots \supseteq \mathcal{F}^k \supseteq \mathcal{F}^{k+1} = 0.$$

(iii) Such that for each $s \in S$, the filtration $\mathcal{F}^\bullet_s$ of $\mathcal{V}_s = V \otimes_{\mathbb{Q}} \mathbb{C}$ defines a pure Hodge structure of weight k on $V = \mathbb{V}_s$.
(iv) Satisfying *Griffiths transversality*:

$$\nabla(\mathcal{F}^p) \subseteq \mathcal{F}^{p-1} \otimes_{\mathcal{O}_S} \Omega^1_S.$$

Remark 4.5. Griffiths transversality is a differential constraint expressing that the Hodge filtration varies "slowly" relative to the flat structure. It is automatically satisfied for VHS arising from geometry (Theorem 4.7).

Definition 4.6. A VHS is *polarized* if it is equipped with a flat bilinear form $Q \colon \mathbb{V} \times \mathbb{V} \to \mathbb{Q}_S$ (i.e., a bilinear form on V invariant under monodromy) that polarizes each fiber.

Theorem 4.7 (Griffiths [21])**.** *Let* $\pi \colon \mathcal{X} \to S$ *be a smooth projective morphism with connected fibers. Then the local system*

$$\mathbb{V} := R^k\pi_*\mathbb{Q}$$

underlies a polarized VHS of weight k*. The Hodge filtration is*

$$\mathcal{F}^p := R^k\pi_*\Omega^{\geq p}_{\mathcal{X}/S}$$

where $\Omega^{\geq p}_{\mathcal{X}/S} := [\Omega^p_{\mathcal{X}/S} \to \Omega^{p+1}_{\mathcal{X}/S} \to \cdots]$ *is the truncated relative de Rham complex.*

Proof. The key points are:

(a) $R^k\pi_*\mathbb{Q}$ is a local system by proper base change.

(b) The relative Hodge-de Rham spectral sequence $E_1^{p,q} = R^q\pi_*\Omega^p_{\mathcal{X}/S} \Rightarrow R^{p+q}\pi_*\Omega^\bullet_{\mathcal{X}/S}$ degenerates at E_1 for smooth projective morphisms.
(c) Griffiths transversality follows from the Gauss-Manin connection being induced by the exterior derivative on relative forms.
(d) Polarization comes from cup product and a relatively ample class.

See [49, 42] for complete proofs. □

4.3. **The period map.**

Definition 4.8. Fix a $\mathbb{Q}$-vector space V of dimension d, a weight k, Hodge numbers $\{h^{p,q}\}_{p+q=k}$, and a polarization form Q. The *period domain* is

$$D := \{F^\bullet \subset V_\mathbb{C} : F^\bullet \text{ defines a } Q\text{-polarized Hodge structure with given Hodge numbers}\}.$$

This is an open subset of the *compact dual* $\check{D}$, which is a flag variety.

Proposition 4.9. *The period domain D is a homogeneous space $D = G_\mathbb{R}/K$ where:*
(i) $G_\mathbb{R} := \mathrm{Aut}(V_\mathbb{R}, Q)$ *is the real Lie group preserving Q.*
(ii) $K \subset G_\mathbb{R}$ *is the compact subgroup preserving a fixed Hodge structure $F_0^\bullet \in D$.*

The compact dual is $\check{D} = G_\mathbb{C}/P$ where P is a parabolic subgroup.

Definition 4.10. Let $(\mathbb{V}, \mathcal{F}^\bullet)$ be a polarized VHS over S. Choosing a basepoint $s_0 \in S$ and identifying $V = \mathbb{V}_{s_0}$, the *period map* is

$$\Phi\colon \tilde{S} \longrightarrow D, \quad \tilde{s} \mapsto \mathcal{F}^\bullet_{\pi(\tilde{s})}$$

where $\tilde{S} \to S$ is the universal cover and we use parallel transport to identify all fibers with V.

The period map descends to $\Phi\colon S \to \Gamma\backslash D$ where $\Gamma = \mathrm{im}(\rho) \subset G_\mathbb{Z}$ is the monodromy group.

Theorem 4.11 (Griffiths)**.** *The period map $\Phi\colon \tilde{S} \to D$ is holomorphic. Griffiths transversality is equivalent to Φ being* horizontal*: $d\Phi(T_{\tilde{s}}\tilde{S}) \subseteq T^{-1,1}_{\Phi(\tilde{s})}D$ where $T^{-1,1}$ is the horizontal tangent bundle.*

4.4. **Monodromy.**

Definition 4.12. The *monodromy group* of a VHS $(\mathbb{V}, \mathcal{F}^\bullet)$ over S is

$$\Gamma := \mathrm{im}(\rho\colon \pi_1(S, s_0) \to \mathrm{GL}(V)).$$

For a polarized VHS, $\Gamma \subset G_\mathbb{Z} := \mathrm{Aut}(V_\mathbb{Z}, Q)$ where $V_\mathbb{Z} \subset V$ is any lattice preserved by monodromy.

Theorem 4.13 (Borel, Monodromy Theorem)**.** *Let $(\mathbb{V}, \mathcal{F}^\bullet)$ be a polarized VHS over a smooth quasi-projective variety S. Then:*
(i) The monodromy group Γ is quasi-unipotent*: every $T \in \Gamma$ has eigenvalues that are roots of unity.*
(ii) More precisely, there exists $m \geq 1$ such that $(T^m - I)^{k+1} = 0$ for all $T \in \Gamma$, where k is the weight.

Proof. This follows from the theory of variations of Hodge structure. The key input is that the period map has finite volume image in $\Gamma\backslash D$, which constrains the monodromy. See [45, Theorem 4.5]. $\square$

Definition 4.14. For a quasi-unipotent transformation T, write $T = T_s T_u$ where T_s is semisimple (with eigenvalues roots of unity) and $T_u = \exp(N)$ is unipotent. The nilpotent operator

$$N := \log(T_u) = (T_u - I) - \frac{1}{2}(T_u - I)^2 + \frac{1}{3}(T_u - I)^3 - \cdots$$

is the *monodromy logarithm*. The series terminates since T_u is unipotent.

5. Degenerations of Hodge structures

We study the behavior of Hodge structures as varieties degenerate, following [45, 46, 9].

5.1. **Semistable degenerations.**

Definition 5.1. A *semistable degeneration* is a proper flat morphism $\pi\colon \mathcal{X} \to \Delta$ from a smooth variety $\mathcal{X}$ to the disk $\Delta = \{t \in \mathbb{C} : |t| < 1\}$ such that:

(i) π is smooth over $\Delta^* := \Delta \setminus \{0\}$.
(ii) The central fiber $X_0 := \pi^{-1}(0)$ is a *reduced* divisor with *simple normal crossings*: locally, $\mathcal{X}$ is isomorphic to $\{(z_1, \ldots, z_{n+1}, t) \in \mathbb{C}^{n+1} \times \Delta : z_1 \cdots z_r = t\}$ for some $r \leq n+1$.

Notation 5.2. For a semistable degeneration, write $X_0 = \bigcup_{i\in I} Y_i$ where each Y_i is an irreducible component. For a nonempty subset $J \subseteq I$, define the stratum

$$Y_J := \bigcap_{j\in J} Y_j.$$

Each Y_J is smooth of dimension $n - |J| + 1$ (when nonempty).

Lemma 5.3. *In a semistable degeneration with* $\dim X_t = n$ *for* $t \neq 0$*:*

(a) Each component Y_i has dimension n.
(b) Each double intersection $Y_{ij} = Y_i \cap Y_j$ (for $i \neq j$) has dimension $n - 1$.
(c) More generally, $\dim Y_J = n - |J| + 1$ *when* $Y_J \neq \emptyset$.

Proof. By the local model, near a point of Y_J, the family is locally $(z_1 \cdots z_r = t)$ with $r = |J|$. The total space $\mathcal{X}$ has dimension $n+1$. The fiber $X_0 = \{t = 0\}$ is the central fiber of dimension n. Within X_0, the component Y_i is $\{z_i = 0\} \cap X_0$, which has dimension n. The intersection Y_J for $|J| = r$ adds $r-1$ additional equations within X_0, so $\dim Y_J = n - (r-1) = n - |J| + 1$. $\square$

Theorem 5.4 (Semistable Reduction)**.** *Let $\pi\colon \mathcal{X} \to \Delta$ be a proper morphism smooth over Δ^*. After a finite base change $\Delta' \to \Delta$, $t \mapsto t^m$, and a birational modification of $\mathcal{X} \times_\Delta \Delta'$, the family becomes semistable.*

Proof. This is a fundamental result in algebraic geometry. See [31, Chapter 2] for the original proof using toroidal embeddings, or [16] for a proof using alterations. $\square$

5.2. The monodromy operator.

Definition 5.5. Let $\pi\colon \mathcal{X} \to \Delta$ be a semistable degeneration. The *monodromy transformation* $T\colon H^k(X_t) \to H^k(X_t)$ is the automorphism induced by parallel transport around a generator of $\pi_1(\Delta^*, t) \cong \mathbb{Z}$.

Theorem 5.6 (Local Monodromy Theorem)**.** *For a semistable degeneration:*

(i) The monodromy T is unipotent: $T = \exp(N)$ for a nilpotent N.
(ii) $N^{k+1} = 0$ where k is the degree of cohomology.
(iii) N is a morphism of type $(-1,-1)$ with respect to the limiting Hodge structure.

Proof. Unipotency follows from the semistability assumption—the Picard-Lefschetz formula shows that T acts by transvections along vanishing cycles. The bound $N^{k+1} = 0$ comes from the monodromy weight filtration. See [45, Theorem 6.1]. □

5.3. Types of degenerations.

Definition 5.7. A semistable degeneration is classified by the nilpotent order of N:

(i) *Type I*: $N = 0$ (smooth family, no degeneration).
(ii) *Type II*: $N \neq 0$ but $N^2 = 0$.
(iii) *Type III* (maximally unipotent): $N^k \neq 0$ but $N^{k+1} = 0$.

More generally, the "type" is determined by the nilpotent orbit $(N_1, \ldots, N_r)$ for multi-parameter degenerations.

Remark 5.8. Type III degenerations are the "most degenerate" and have the richest boundary structure. For a VHS of weight k on a family of n-dimensional varieties, Type III means $N^n \neq 0$ (for middle cohomology $k = n$). The central fiber has the maximal number of components.

Proposition 5.9. *A semistable degeneration of n-dimensional varieties is Type III if and only if:*

(a) The dual graph of $X_0 = \bigcup Y_i$ is "maximally connected."
(b) The limiting mixed Hodge structure has weights spread across $[0, 2n]$.
(c) The monodromy weight filtration $W_\bullet(N, n)$ has $\mathrm{Gr}^W_{2n} \neq 0$ and $\mathrm{Gr}^W_0 \neq 0$.

6. Limiting mixed Hodge structures

We construct the limiting mixed Hodge structure following Schmid [45] and Steenbrink [46].

6.1. Mixed Hodge structures.

Definition 6.1. A *mixed Hodge structure (MHS)* on a $\mathbb{Q}$-vector space V consists of:

(i) An increasing *weight filtration* $W_\bullet$ on V:

$$0 = W_{m-1} \subseteq W_m \subseteq W_{m+1} \subseteq \cdots \subseteq W_M = V.$$

(ii) A decreasing *Hodge filtration* $F^\bullet$ on $V_\mathbb{C}$:

$$V_\mathbb{C} = F^0 \supseteq F^1 \supseteq \cdots \supseteq F^n \supseteq F^{n+1} = 0.$$

(iii) Such that $F^\bullet$ induces a pure Hodge structure of weight k on each graded piece:

$$\mathrm{Gr}_k^W V := W_k/W_{k-1}.$$

Definition 6.2. A *morphism* of MHS $\phi\colon V \to W$ is a $\mathbb{Q}$-linear map preserving both filtrations: $\phi(W_kV) \subseteq W_kW$ and $\phi_{\mathbb{C}}(F^pV_{\mathbb{C}}) \subseteq F^pW_{\mathbb{C}}$.

Theorem 6.3 (Deligne [11])**.** *Every morphism of MHS is strict with respect to both filtrations. The category of MHS is abelian.*

Proof. Strictness means $\phi(W_kV) = W_kW \cap \mathrm{im}(\phi)$ and similarly for $F^\bullet$. The proof uses the existence of a canonical splitting of MHS. See [42, §2.3]. □

6.2. The monodromy weight filtration.

Theorem 6.4. *Let V be a finite-dimensional $\mathbb{Q}$-vector space, $N\colon V \to V$ a nilpotent endomorphism, and $k \in \mathbb{Z}$. There exists a unique increasing filtration $W_\bullet = W_\bullet(N,k)$, called the* monodromy weight filtration centered at k*, satisfying:*

(i) $N(W_\ell) \subseteq W_{\ell-2}$ *for all* ℓ.
(ii) For each $\ell \geq 0$*, the map* $N^\ell\colon \mathrm{Gr}_{k+\ell}^W V \to \mathrm{Gr}_{k-\ell}^W V$ *is an isomorphism.*

Proof. **Existence:** We construct $W_\bullet$ inductively. Since N is nilpotent, let r be the largest integer with $N^r \neq 0$.

For $r = 0$ (i.e., $N = 0$): set $W_\ell = V$ for $\ell \geq k$ and $W_\ell = 0$ for $\ell < k$.

For $r \geq 1$: Consider the exact sequence

$$0 \to \ker(N) \to V \xrightarrow{N} \mathrm{im}(N) \to 0.$$

By induction on nilpotent order, $\mathrm{im}(N)$ admits a monodromy weight filtration $W'_\bullet(N|_{\mathrm{im}(N)}, k-2)$ (the restriction of N has lower nilpotent order).

Define:

$$\begin{aligned} W_\ell &:= N^{-1}(W'_{\ell-2}) \cap \ker(N^{(k+\ell+2)/2}) \quad \text{for } \ell < k, \\ W_k &:= \ker(N), \\ W_\ell &:= W_k + N^{-1}(W'_{\ell-2}) \quad \text{for } \ell > k. \end{aligned}$$

Uniqueness: Suppose $W_\bullet$ and $W'_\bullet$ both satisfy (i) and (ii). We show $W_\ell = W'_\ell$ by induction.

From (i), $N(W_k) \subseteq W_{k-2}$. From (ii) with $\ell = 0$, $N^0 = \mathrm{id}$ is an “isomorphism” on Gr_k^W, meaning $\mathrm{Gr}_k^W = \ker(N|_{\mathrm{Gr}_k^W})$. This forces $W_k = \ker(N)$.

Similarly $W'_k = \ker(N)$, so $W_k = W'_k$.

Inductively, using the isomorphism $N^\ell\colon \mathrm{Gr}_{k+\ell}^W \to \mathrm{Gr}_{k-\ell}^W$, one shows $W_{k\pm\ell} = W'_{k\pm\ell}$. □

Corollary 6.5. *For the monodromy weight filtration $W_\bullet = W_\bullet(N,k)$:*

(a) $\ker(N) = W_k$.
(b) $\mathrm{im}(N) \subseteq W_{k-2}$.
(c) $\ker(N^{\ell+1}) = W_{k+\ell}$ *for all* $\ell \geq 0$.

Proof. (a) From the proof of uniqueness above.

(b) From property (i): $\mathrm{im}(N) = N(V) = N(W_M) \subseteq W_{M-2}$, and iterating, $\mathrm{im}(N) \subseteq W_{k-2}$.

(c) The isomorphism $N^{\ell+1}\colon \mathrm{Gr}^W_{k+\ell+1} \to \mathrm{Gr}^W_{k-\ell-1}$ implies $\ker(N^{\ell+1}) \cap \mathrm{Gr}^W_{k+\ell+1} = 0$. Thus elements of $\ker(N^{\ell+1})$ lie in $W_{k+\ell}$. Conversely, for $v \in W_{k+\ell}$, we have $N^{\ell+1}(v) \in W_{k-\ell-2}$ by (i), and the isomorphism property shows this forces $N^{\ell+1}(v) = 0$. □

6.3. **The limiting mixed Hodge structure.** Let $(\mathbb{V}, \mathcal{F}^\bullet)$ be a polarized VHS of weight k over Δ^*. Let $T = \exp(N)$ be the (unipotent) monodromy.

Definition 6.6. The *nilpotent orbit approximation* to the period map $\Phi\colon \mathbb{H} \to D$ (where $\mathbb{H}$ is the upper half-plane, covering Δ^*) is:

$$\Phi_\infty(z) := \exp(zN) \cdot F^\bullet_\infty$$

where $F^\bullet_\infty := \lim_{\Im(z)\to\infty} \exp(-zN) \cdot \Phi(z)$.

Theorem 6.7 (Schmid [45])**.** *Let $(\mathbb{V}, \mathcal{F}^\bullet)$ be a polarized VHS of weight k over Δ^*. Then:*

(i) The limit $F^\bullet_\infty := \lim_{t\to 0} \exp\left(-\frac{\log t}{2\pi i} N\right) \mathcal{F}^\bullet_t$ exists.
(ii) $(V, W_\bullet(N,k), F^\bullet_\infty)$ is a mixed Hodge structure.
(iii) $N\colon V \to V(-1,-1)$ is a morphism of MHS (i.e., N shifts weights by -2 and Hodge type by $(-1,-1)$).
(iv) The MHS is $\mathbb{R}$-split: there exists a bigrading $V_\mathbb{C} = \bigoplus I^{p,q}$ with $F^p = \bigoplus_{a\geq p} I^{a,b}$ and $W_\ell = \bigoplus_{a+b\leq \ell} I^{a,b}$.

Proof. The proof involves detailed analysis of the asymptotic behavior of the period map. See [45, Theorem 6.16] and [42, §11]. □

Definition 6.8. The MHS $(V, W_\bullet(N,k), F^\bullet_\infty)$ is called the *limiting mixed Hodge structure (LMHS).*

7. The Clemens-Schmid exact sequence

The Clemens-Schmid sequence relates the cohomology of degenerating fibers to that of the central fiber.

7.1. **Statement of the exact sequence.** Let $\pi\colon \mathcal{X} \to \Delta$ be a semistable degeneration with smooth fibers X_t for $t \neq 0$ and central fiber $X_0 = \bigcup_{i\in I} Y_i$.

Theorem 7.1 (Clemens-Schmid [10, 45])**.** *There is an exact sequence of mixed Hodge structures:*

$$\cdots \to H^{k-2}(X_0)(-1) \xrightarrow{\delta} H^k(X_0) \xrightarrow{i^*} H^k_{\lim}(X_t) \xrightarrow{N} H^k_{\lim}(X_t)(-1) \xrightarrow{\delta} H^{k+1}(X_0) \to \cdots$$

where:

(i) $H^k(X_0)$ carries the canonical MHS on a normal crossing variety [12].
(ii) $H^k_{\lim}(X_t)$ denotes cohomology with the LMHS.
(iii) $i^*\colon H^k(X_0) \to H^k_{\lim}(X_t)$ is the specialization map.
(iv) N is the monodromy logarithm.
(v) δ is a connecting homomorphism.
(vi) All maps are morphisms of MHS.

Remark 7.2. The sequence is *not* short exact in general. The maps δ measure the "defect" of the specialization and monodromy.

7.2. **The key corollary.**

Corollary 7.3. *Let $\alpha \in H^k_{\lim}(X_t)$. If $N(\alpha) = 0$, then there exists $\beta \in H^k(X_0, \mathbb{Q})$ with $i^*(\beta) = \alpha$.*

Proof. Exactness at $H^k_{\lim}(X_t)$ gives $\ker(N) = \operatorname{im}(i^*)$. Thus $N(\alpha) = 0$ implies $\alpha \in \operatorname{im}(i^*)$. □

Remark 7.4. Corollary 7.3 shows that monodromy-invariant classes on smooth fibers *descend* to the central fiber. The condition $N(\alpha) = 0$ is guaranteed by the stabilizer cover construction (§15).

7.3. **The MHS on a normal crossing variety.**

Definition 7.5. Let $X_0 = \bigcup_{i \in I} Y_i$ be a reduced simple normal crossing variety. The MHS on $H^k(X_0)$ is constructed via the *weight spectral sequence*:

$$E_1^{-r,k+r} = \bigoplus_{|J|=r+1} H^{k-2r}(Y_J)(-r) \Longrightarrow H^k(X_0).$$

Proposition 7.6. *The weight spectral sequence has:*

(a) E_1-differential $d_1 \colon E_1^{-r,k+r} \to E_1^{-r+1,k+r}$ given by alternating restriction and Gysin maps.
(b) The filtration on the abutment is the weight filtration: $\operatorname{Gr}^W_{k+r} H^k(X_0) = E_\infty^{-r,k+r}$.
(c) Each $E_1^{-r,k+r}$ is a direct sum of pure Hodge structures of weight $k - 2r + 2r = k$ (after Tate twist).

Proof. The construction uses the Mayer-Vietoris spectral sequence for the covering $X_0 = \bigcup Y_i$, combined with the weight filtration from Deligne's theory of MHS on singular varieties. See [12, 46, 42]. □

7.4. **Restriction to strata.**

Proposition 7.7. *Let $\beta \in H^{2p}(X_0)$ be a class with $i^*(\beta) = \alpha \in \operatorname{Hdg}^p(X_t)$ (a Hodge class on smooth fibers). Then for each stratum Y_J, the restriction $\beta_J := \beta|_{Y_J}$ is a Hodge class on Y_J of* ***codimension*** *p (not $p - |J| + 1$).*

Proof. Restriction preserves cohomological degree: $\beta \in H^{2p}(X_0)$ restricts to $\beta_J \in H^{2p}(Y_J)$. On Y_J of dimension $\dim Y_J = n - |J| + 1$, a class in H^{2p} corresponds to codimension p.

For this to be nonzero, we need $2p \le 2 \dim Y_J = 2(n - |J| + 1)$, i.e., $|J| \le n - p + 1$.

The Hodge type is preserved: since α is of type (p, p) and all maps in Clemens-Schmid are MHS morphisms, β_J is of type (p, p), hence $\beta_J \in \operatorname{Hdg}^p(Y_J)$. □

Corollary 7.8. *For $|J| \ge 2$, the stratum Y_J has $\dim Y_J = n - |J| + 1 \le n - 1 < n$. The Hodge class $\beta_J \in \operatorname{Hdg}^p(Y_J)$ is covered by the induction hypothesis, provided $p \le \dim Y_J$ (equivalently, $|J| \le n - p + 1$).*

Remark 7.9. For strata Y_J with $|J| > n - p + 1$, we have $\dim Y_J < p$, so $H^{2p}(Y_J) = 0$ and $\beta_J = 0$ automatically. Thus only strata with $|J| \le n - p + 1$ contribute nontrivially.

7.5. **Proof of Clemens-Schmid.**

Proof sketch of Theorem 7.1. The key steps are:

Step 1: Construct a *relative log de Rham complex* $\Omega^\bullet_{\mathcal{X}/\Delta}(\log X_0)$ that computes the cohomology of smooth fibers with logarithmic singularities.

Step 2: Show that the nearby cycles $\psi_t(\mathbb{Q}_{\mathcal{X}})$ satisfy a distinguished triangle:

$$\mathbb{Q}_{X_0} \to \psi_t\mathbb{Q}_{\mathcal{X}} \xrightarrow{N} \psi_t\mathbb{Q}_{\mathcal{X}}(-1) \xrightarrow{+1} .$$

Step 3: Take hypercohomology to get the exact sequence:

$$H^k(X_0) \to H^k(\psi_t\mathbb{Q}_{\mathcal{X}}) \xrightarrow{N} H^k(\psi_t\mathbb{Q}_{\mathcal{X}})(-1) \to H^{k+1}(X_0).$$

Step 4: Identify $H^k(\psi_t\mathbb{Q}_{\mathcal{X}})$ with $H^k_{\lim}(X_t)$ via the comparison theorem.

Step 5: Verify that all maps are morphisms of MHS.

See [41, 42] for complete details. □

8. Algebraic cycles and Chow groups

We review the theory of algebraic cycles following [18].

8.1. **Cycles and rational equivalence.**

Definition 8.1. Let X be a variety of dimension n. A *cycle of codimension* p (or *dimension* $n-p$) on X is a formal $\mathbb{Z}$-linear combination

$$Z = \sum_i n_i[V_i]$$

where each $V_i \subset X$ is an irreducible subvariety of codimension p and $n_i \in \mathbb{Z}$. The group of codimension-p cycles is denoted $Z^p(X)$.

Definition 8.2. Two cycles $Z_1, Z_2 \in Z^p(X)$ are *rationally equivalent*, written $Z_1 \sim_{\text{rat}} Z_2$, if there exist:

(i) Irreducible subvarieties $W_j \subset X \times \mathbb{P}^1$ of codimension p, flat over $\mathbb{P}^1$.
(ii) Such that $Z_1 - Z_2 = \sum_j([W_j]_0 - [W_j]_\infty)$.

Here $[W_j]_t := W_j \cap (X \times \{t\})$ denotes the fiber over $t \in \mathbb{P}^1$.

Definition 8.3. The *Chow group* of codimension-p cycles is

$$\mathrm{CH}^p(X) := Z^p(X)/\sim_{\text{rat}} .$$

We write $\mathrm{CH}^p(X)_{\mathbb{Q}} := \mathrm{CH}^p(X) \otimes_{\mathbb{Z}} \mathbb{Q}$ for the Chow group with rational coefficients.

Proposition 8.4 (Basic Properties)**.** *Let X be a smooth variety of dimension n.*

(a) $\mathrm{CH}^0(X) = \mathbb{Z}$ *(generated by $[X]$).*
(b) $\mathrm{CH}^n(X) = \mathbb{Z}^{\oplus\#(\textit{closed points up to rational equiv})}$.
(c) $\mathrm{CH}^1(X) \cong \mathrm{Pic}(X)$ *(divisor classes modulo linear equivalence).*
(d) There is an intersection product $\mathrm{CH}^p(X)\times\mathrm{CH}^q(X) \to \mathrm{CH}^{p+q}(X)$ *making* $\mathrm{CH}^*(X)$ *a commutative graded ring.*

8.2. **The cycle class map.**

Theorem 8.5. *For a smooth variety X, there is a natural homomorphism*

$$\mathrm{cl}\colon \mathrm{CH}^p(X) \longrightarrow H^{2p}(X, \mathbb{Z})$$

called the cycle class map*.*

Construction 8.6. For an irreducible subvariety $V \subset X$ of codimension p:

(i) Let $\tilde{V} \to V$ be a resolution of singularities.
(ii) The fundamental class $[\tilde{V}] \in H_{2(n-p)}(\tilde{V}, \mathbb{Z})$ pushes forward to $H_{2(n-p)}(X, \mathbb{Z})$.
(iii) By Poincaré duality, this defines $[V] \in H^{2p}(X, \mathbb{Z})$.

Extend by linearity to all cycles. Rational equivalence maps to zero (shown using the homotopy invariance of cohomology).

Proposition 8.7. *The cycle class map satisfies:*

(a) Multiplicativity*:* $\mathrm{cl}(Z \cdot W) = \mathrm{cl}(Z) \cup \mathrm{cl}(W)$ *for properly intersecting cycles.*
(b) First Chern class*: For a Cartier divisor D,* $\mathrm{cl}(D) = c_1(\mathcal{O}_X(D))$*.*
(c) Pushforward*: For $f\colon X \to Y$ proper,* $\mathrm{cl}(f_*Z) = f_* \,\mathrm{cl}(Z)$*.*
(d) Pullback*: For $f\colon X \to Y$ flat,* $\mathrm{cl}(f^*Z) = f^* \,\mathrm{cl}(Z)$*.*

Proof. See [18, Chapter 19]. □

Proposition 8.8. *For X smooth projective, the image of* cl *lies in Hodge classes:*

$$\mathrm{im}(\mathrm{cl}\colon \mathrm{CH}^p(X)_{\mathbb{Q}} \to H^{2p}(X, \mathbb{Q})) \subseteq \mathrm{Hdg}^p(X).$$

Proof. For an irreducible subvariety $V \subset X$ of codimension p, the class $[V]$ is Poincaré dual to integration over V. For a $(2n-2p)$-form ω, $\int_X [V] \wedge \omega = \int_V \omega|_V$.

The integral $\int_V \omega|_V$ vanishes unless $\omega|_V$ has a $(n-p, n-p)$-component, which requires ω to have at least a (p,p)-component. Thus $[V] \in H^{p,p}(X)$.

The general case follows by linearity and the observation that rational equivalences also preserve Hodge type (by the same argument applied to the universal family over $\mathbb{P}^1$). □

8.3. **Other equivalence relations.**

Definition 8.9. For cycles on a smooth projective X:

(i) *Homological equivalence*: $Z \sim_{\mathrm{hom}} 0$ if $\mathrm{cl}(Z) = 0$ in $H^{2p}(X, \mathbb{Q})$.
(ii) *Numerical equivalence*: $Z \sim_{\mathrm{num}} 0$ if $\deg(Z \cdot W) = 0$ for all $W \in \mathrm{CH}^{n-p}(X)_{\mathbb{Q}}$.

Proposition 8.10. *We have inclusions:*

$$\sim_{\mathrm{rat}} \subseteq \sim_{\mathrm{hom}} \subseteq \sim_{\mathrm{num}} .$$

Proof. The first inclusion follows from homotopy invariance of cohomology. The second follows from the fact that $\mathrm{cl}(Z) \cup \mathrm{cl}(W) = \mathrm{cl}(Z \cdot W)$ and the degree map $\deg\colon H^{2n}(X) \to \mathbb{Q}$ factors through the cycle class. □

Remark 8.11. Grothendieck's Standard Conjecture D asserts that $\sim_{\mathrm{hom}} = \sim_{\mathrm{num}}$. This will follow from Theorem 1.3 (see Section 27).

9. The Chow variety and properness

The properness of the Chow variety is essential for extending cycles across degenerations.

9.1. **The Chow variety.**

Definition 9.1. Let $X \subset \mathbb{P}^N$ be a projective variety. The *Chow variety* $\mathrm{Chow}_{d,e}(X)$ parametrizes effective cycles on X of dimension d and degree e (with respect to the embedding).

Theorem 9.2 (Chow-van der Waerden). *For $X \subset \mathbb{P}^N$ projective, the Chow variety $\mathrm{Chow}_{d,e}(X)$ exists as a projective variety.*

Proof sketch. Cycles of dimension d and degree e are parametrized by their *Chow forms*: homogeneous polynomials in the Plücker coordinates of $(N-d-1)$-planes. The Chow variety is cut out by incidence conditions. See [20, §I.3]. □

Definition 9.3. Let $\pi\colon \mathcal{X} \to S$ be a projective morphism. The *relative Chow variety* $\mathrm{Chow}^p(\mathcal{X}/S) \to S$ is the scheme whose fiber over $s \in S$ parametrizes codimension-p cycles on X_s.

9.2. **Barlet's properness theorem.**

Theorem 9.4 (Barlet [3]). *Let $\pi\colon \mathcal{X} \to S$ be a projective morphism of complex analytic spaces. The relative Chow variety $\mathrm{Chow}^p(\mathcal{X}/S) \to S$ is proper.*

Remark 9.5. Barlet's theorem is stated for complex analytic spaces, which includes the algebraic case by GAGA. The properness is in the sense of complex analytic geometry: every sequence has a convergent subsequence.

Corollary 9.6 (Cycle Extension). *Let $\pi\colon \mathcal{X} \to S$ be projective with S irreducible. Let $U \subseteq S$ be a dense open subset, and let $\{Z_s\}_{s\in U}$ be a flat family of cycles on fibers of $\mathcal{X}|_U \to U$.*

Then this family extends uniquely to a flat family $\{Z_s\}_{s\in S}$ over all of S.

Proof. The family over U defines a morphism $\phi\colon U \to \mathrm{Chow}^p(\mathcal{X}/S)$. The graph $\Gamma_\phi \subset U \times_S \mathrm{Chow}^p(\mathcal{X}/S)$ has closure $\bar{\Gamma}_\phi \subset S \times_S \mathrm{Chow}^p(\mathcal{X}/S)$.

By properness of $\mathrm{Chow}^p(\mathcal{X}/S) \to S$, the projection $\bar{\Gamma}_\phi \to S$ is proper. Since U is dense and S is irreducible, $\bar{\Gamma}_\phi \to S$ is surjective.

For any $s \in S$, the fiber $(\bar{\Gamma}_\phi)_s$ is nonempty, giving a limit cycle Z_s. Uniqueness follows from the separatedness of $\mathrm{Chow}^p(\mathcal{X}/S)$. □

Remark 9.7. Critical observation: Theorem 9.4 asserts *extension* of existing cycle families. It does NOT assert *creation* of cycles from cohomological data. That is:

- If you have cycles Z_t on smooth fibers X_t, properness extends them to the central fiber X_0.
- If you have a Hodge class α_t on smooth fibers, properness does NOT produce a cycle with $\mathrm{cl}(Z_t) = \alpha_t$.

The Hodge conjecture is precisely the statement that such cycles exist. Our proof constructs them using the induction hypothesis and the cycle-Lefschetz theorem.

9.3. **Specialization of cycles.**

Definition 9.8. Let $\pi\colon \mathcal{X} \to \Delta$ be a flat projective family over the disk with reduced central fiber X_0. The *specialization map*

$$\mathrm{sp}\colon\ \mathrm{CH}^p(X_\eta)_{\mathbb{Q}} \longrightarrow \mathrm{CH}^p(X_0)_{\mathbb{Q}}$$

sends a cycle Z on the generic fiber to the central fiber of its flat closure in $\mathcal{X}$.

Proposition 9.9. *If X_0 is reduced, the specialization map is surjective.*

Proof. Let Z_0 be a cycle on X_0. By properness of the Chow variety, Z_0 is a limit of cycles on nearby smooth fibers X_t for $t \neq 0$ small. Taking such a cycle as the "spreading" of Z_0, we find $\mathrm{sp}(Z) = Z_0$.

Formally: the inclusion $X_0 \hookrightarrow \mathcal{X}$ defines $\mathrm{sp}^{-1}(Z_0) \neq \emptyset$ by properness. □

9.4. **Continuity of cycle classes.**

Proposition 9.10. *Let $\pi\colon \mathcal{X} \to S$ be a smooth projective morphism. The cycle class map*

$$\mathrm{cl}\colon\ \mathrm{Chow}^p(\mathcal{X}/S) \longrightarrow R^{2p}\pi_*\mathbb{Q}$$

is continuous, where $R^{2p}\pi_*\mathbb{Q}$ is viewed as a local system.

Proof. The cycle class is computed by integration, which varies continuously (in fact, holomorphically) in families. □

Corollary 9.11. *Let $\{Z_s\}_{s\in S}$ be a flat family of cycles and α a flat section of $R^{2p}\pi_*\mathbb{Q}$. If $\mathrm{cl}(Z_{s_0}) = \alpha_{s_0}$ at some point, then $\mathrm{cl}(Z_s) = \alpha_s$ for all $s \in S$ (in the same connected component).*

Proof. Both $s \mapsto \mathrm{cl}(Z_s)$ and $s \mapsto \alpha_s$ are continuous sections of the local system $R^{2p}\pi_*\mathbb{Q}$. They agree at s_0. Since the fibers of a local system are discrete, continuous sections that agree at a point agree everywhere on a connected set. □

10. The cohomological Lefschetz theorem

We review the classical Lefschetz hyperplane theorem for cohomology.

10.1. **Statement.**

Theorem 10.1 (Lefschetz Hyperplane Theorem)**.** *Let Y be a smooth projective variety of dimension m, and let $D \subset Y$ be a smooth hypersurface cut out by a section of an ample line bundle $\mathcal{L}$. Then:*

(i) The restriction map $H^k(Y,\mathbb{Q}) \to H^k(D,\mathbb{Q})$ is an isomorphism for $k < m-1$.
(ii) The restriction map $H^{m-1}(Y,\mathbb{Q}) \to H^{m-1}(D,\mathbb{Q})$ is injective.

Dually:

(i') The Gysin map $H^k(D,\mathbb{Q}) \to H^{k+2}(Y,\mathbb{Q})$ is an isomorphism for $k > m-1$.
(ii') The Gysin map $H^{m-1}(D,\mathbb{Q}) \to H^{m+1}(Y,\mathbb{Q})$ is surjective.

Proof. We sketch the proof using Morse theory.

Step 1: Choose a generic pencil in $|\mathcal{L}|$ with base locus $B \subset Y$ of codimension 2. Blowing up B gives $\tilde{Y} \to \mathbb{P}^1$ with general fiber isomorphic to D.

Step 2: The function $f := -\log|s|^2$ where s is a section defining D is a Morse function on $Y \setminus D$. The critical points correspond to singular fibers of the pencil.

Step 3: By Morse theory, Y is built from D by attaching cells of dimension $\geq m$ (since critical points have Morse index $\geq m$ by the ampleness condition).

Step 4: The cellular structure gives $H_k(Y, D) = 0$ for $k < m$, which translates to the stated cohomological properties via the long exact sequence.

See [49, Chapter 7] or [17, §1.2] for complete details. □

10.2. **Consequences for Hodge classes.**

Corollary 10.2. *Let Y be smooth projective of dimension m, and $D \subset Y$ a smooth ample divisor. For $2p \leq m-1$, the restriction*

$$H^{2p}(Y, \mathbb{Q}) \longrightarrow H^{2p}(D, \mathbb{Q})$$

is injective. In particular, a Hodge class $\beta \in \mathrm{Hdg}^p(Y)$ is determined by its restriction to D.

Proof. For $2p < m-1$, Theorem 10.1(i) gives an isomorphism. For $2p = m-1$ (which requires m odd), part (ii) gives injectivity. □

Corollary 10.3. *Under restriction $H^{2p}(Y) \to H^{2p}(D)$:*

(a) Hodge type is preserved: $H^{p,p}(Y) \to H^{p,p}(D)$.
(b) Hodge classes restrict to Hodge classes: $\mathrm{Hdg}^p(Y) \to \mathrm{Hdg}^p(D)$.

Proof. Restriction of differential forms preserves type. For (b), if $\alpha \in H^{2p}(Y, \mathbb{Q}) \cap H^{p,p}(Y)$, then $\alpha|_D \in H^{2p}(D, \mathbb{Q}) \cap H^{p,p}(D)$. □

10.3. **The affine Lefschetz theorem.**

Theorem 10.4 (Affine Lefschetz)**.** *Let Y be smooth projective of dimension m, and $D \subset Y$ an ample divisor. The complement $U := Y \setminus D$ has:*

(i) $H^k(U, \mathbb{Q}) = 0$ for $k > m$.
(ii) $H^k(U, \mathbb{Q}) \to H^k(Y, \mathbb{Q})$ is an isomorphism for $k < m$.

Proof. By Morse theory, U has the homotopy type of a CW complex of dimension $\leq m$. The map to Y is m-connected. □

10.4. **Lefschetz for pairs.**

Theorem 10.5 (Lefschetz for Pairs)**.** *Let Y be smooth projective of dimension m, and let $D \subset Y$ be a smooth ample divisor. The pair (Y, D) is $(m-1)$-connected:*

$$H^k(Y, D; \mathbb{Q}) = 0 \quad \textit{for } k < m.$$

Equivalently, the Gysin map $\iota_* : H^{k-2}(D) \to H^k(Y)$ is surjective for $k < m$.

Proof. By the Morse-theoretic analysis of Theorem 10.1, the space Y is obtained from D by attaching cells of dimension $\geq m$. Thus (Y, D) has no relative homology in degrees $< m$.

By universal coefficients, $H^k(Y, D; \mathbb{Q}) \cong \mathrm{Hom}(H_k(Y, D; \mathbb{Z}), \mathbb{Q}) = 0$ for $k < m$.

The equivalence with Gysin surjectivity follows from the long exact sequence:

$$\cdots \to H^{k-2}(D) \xrightarrow{\iota_*} H^k(Y) \to H^k(Y, D) \to \cdots$$

If $H^k(Y, D) = 0$, then ι_* is surjective onto $H^k(Y)$. □

Corollary 10.6. *For smooth projective Y of dimension m and smooth ample divisor D:*

$$\iota_* : H^{2p-2}(D) \twoheadrightarrow H^{2p}(Y) \quad \textit{for } 2p < m.$$

Remark 10.7. The Lefschetz theorems encode the principle that "ample divisors capture most of the topology." For our purposes:

- Cohomological Lefschetz: Classes on Y are determined by restrictions to D (for $2p \leq m - 1$).
- Cycle-theoretic Lefschetz (§11): Cycles on D lift to cycles on Y (for $p \leq m - 2$).

The combination creates and identifies cycles.

11. The cycle-theoretic weak Lefschetz theorem

We establish the key theorem for lifting algebraicity from divisors.

11.1. **Gysin compatibility.**

The following standard result is the foundation of the argument.

Proposition 11.1 (Gysin-cycle compatibility [18, Theorem 6.2])**.** *Let $\iota : D \hookrightarrow Y$ be a regular embedding of smooth varieties. The Gysin pushforward on Chow groups*

$$\iota_* : \mathrm{CH}^{p-1}(D)_{\mathbb{Q}} \longrightarrow \mathrm{CH}^p(Y)_{\mathbb{Q}}$$

is compatible with the cycle class map:

$$\mathrm{cl}_Y(\iota_*(W)) = \iota_*(\mathrm{cl}_D(W)) \quad \textit{for all } W \in \mathrm{CH}^{p-1}(D)_{\mathbb{Q}}.$$

Proof. The Gysin pushforward ι_* on Chow groups is defined via the deformation to the normal cone construction [18, Chapter 6]. The compatibility with cycle class follows from the fact that both the Chow-theoretic and cohomological Gysin maps are defined by the same geometric construction (cap product with the fundamental class of D in Y). See [18, Proposition 19.2] for the compatibility statement. □

Remark 11.2 (Direction of the argument)**.** The proposition states: *algebraic cycles push forward to algebraic cycles.* This is trivial since $\iota_* : \mathrm{CH}^* \to \mathrm{CH}^*$ is well-defined.

What is *not* claimed: that Hodge classes in the image of Gysin on cohomology are algebraic. That would be circular. The argument below carefully avoids this: we find a Hodge preimage γ, prove γ is algebraic by induction, then push forward the representing cycle.

11.2. Statement.

Theorem 11.3 (Cycle Extension). *Let Y be smooth projective of dimension m, $D \subset Y$ a smooth ample divisor. Let $\beta \in \mathrm{Hdg}^p(Y)$ with $2p < m - 1$.*

Assume:

(H1) The Hodge conjecture holds for smooth varieties of dimension $< m$.

Then: *β is algebraic.*

Remark 11.4 (Hypothesis (H2) removed). The original statement included "(H2): $\beta|_D$ is algebraic." This is redundant: (H1) implies every Hodge class on D is algebraic, including $\beta|_D$.

11.3. Proof.

Proof. The proof has three parts: (A) show β lies in the Gysin image, (B) identify the Gysin preimage as Hodge, (C) use (H1) to get algebraicity.

Part A: Cohomological Gysin surjectivity.

Consider the long exact sequence for the pair (Y, D):

$$\cdots \to H^{2p-2}(D) \xrightarrow{\iota_*} H^{2p}(Y) \xrightarrow{j^*} H^{2p}(Y, D) \to \cdots$$

Claim: $H^{2p}(Y, D; \mathbb{Q}) = 0$.

Proof of claim: By the Lefschetz hyperplane theorem for pairs (Theorem 10.5), the pair (Y, D) is $(m-1)$-connected when D is a smooth ample divisor in smooth projective Y of dimension m. This means:

$$H^k(Y, D; \mathbb{Q}) = 0 \quad \text{for } k < m.$$

Since $2p < m - 1 < m$, we have $H^{2p}(Y, D; \mathbb{Q}) = 0$. $\square_{\text{claim}}$

Consequence: The Gysin map $\iota_* : H^{2p-2}(D) \to H^{2p}(Y)$ is surjective.

Therefore $\beta = \iota_*(\gamma)$ for some $\gamma \in H^{2p-2}(D)$.

Part B: The preimage is Hodge.

The Gysin map ι_* is a morphism of Hodge structures of type $(1, 1)$.

Since $\beta \in H^{p,p}(Y)$, we have:

$$\iota_*(\gamma) = \beta \in H^{p,p}(Y) \implies \gamma \in H^{p-1,p-1}(D).$$

Therefore $\gamma \in \mathrm{Hdg}^{p-1}(D)$.

Part C: Algebraicity via induction.

By hypothesis (H1), the Hodge conjecture holds for D (since $\dim D = m - 1 < m$).

Therefore $\gamma = \mathrm{cl}_D(W)$ for some $W \in \mathrm{CH}^{p-1}(D)_{\mathbb{Q}}$.

Conclusion.

Define $Z := \iota_*(W) \in \mathrm{CH}^p(Y)_{\mathbb{Q}}$.

By Proposition 11.1 (Gysin-cycle compatibility):

$$\mathrm{cl}_Y(Z) = \mathrm{cl}_Y(\iota_*(W)) = \iota_*(\mathrm{cl}_D(W)) = \iota_*(\gamma) = \beta.$$

Therefore β is algebraic. $\square$

11.4. Clarification: what the proof does NOT use.

Remark 11.5. The proof uses:

(1) Lefschetz for pairs (Theorem 10.5) to get Gysin surjectivity on H^*.
(2) The induction hypothesis (to get γ algebraic on D).
(3) Compatibility of $\mathrm{cl} : \mathrm{CH}^* \to H^*$ with Gysin (Proposition 11.1).

The proof does NOT use:

(a) Any claim that $\mathrm{CH}^p(U) = 0$ for affine U. (False in general; see Mumford.)
(b) Any "Lefschetz theorem for Chow groups." (No such theorem is known.)
(c) Any relation between W (representing $\beta|_D$) and W' (representing γ).

Regarding (c): The hypothesis gives $\beta|_D = \mathrm{cl}(W)$ for some cycle W. The proof finds γ with $\iota_*(\gamma) = \beta$, then finds W' with $\mathrm{cl}(W') = \gamma$ via (H1). These cycles W and W' are generally different: $W \in \mathrm{CH}^p(D)$ while $W' \in \mathrm{CH}^{p-1}(D)$, and $\mathrm{cl}(W) = L_D \cdot \mathrm{cl}(W')$ where $L_D = c_1(N_{D/Y})$.

11.5. The localization sequence (for reference).

Theorem 11.6 (Localization)**.** *For Y smooth, $D \subset Y$ a smooth divisor, $U := Y \setminus D$:*

$$\mathrm{CH}^{p-1}(D)_{\mathbb{Q}} \xrightarrow{\iota_*} \mathrm{CH}^p(Y)_{\mathbb{Q}} \xrightarrow{j^*} \mathrm{CH}^p(U)_{\mathbb{Q}} \to 0.$$

Proof. See [18, Proposition 1.8]. □

Remark 11.7. The localization sequence shows $\mathrm{CH}^p(Y)_{\mathbb{Q}}$ is an extension of $\mathrm{CH}^p(U)_{\mathbb{Q}}$ by $\mathrm{im}(\iota_*)$. It does NOT imply $\mathrm{CH}^p(U) = 0$.

12. The cycle extension lemma

We establish the key lemma for lifting algebraicity from divisors to ambient varieties.

12.1. The main extension result.

Lemma 12.1 (Cycle Extension)**.** *Assume the Hodge conjecture for smooth projective varieties of dimension $< n$. Let Y be smooth projective of dimension $m \leq n$, and let $D \subset Y$ be a smooth ample divisor.*

Let $\beta \in \mathrm{Hdg}^p(Y)$ be a Hodge class such that $\beta|_D$ is algebraic. If $2p < m - 1$, then β is algebraic.

Proof. **Step 1: Cohomological setup.**

By the Lefschetz hyperplane theorem, for $2p < m - 1 = \dim D$:

$$H^{2p}(Y, \mathbb{Q}) \xrightarrow{\mathrm{res}} H^{2p}(D, \mathbb{Q}) \text{ is injective.}$$

Step 2: Gysin surjectivity (cohomological).

Since D is ample and $2p < m$, the Gysin map

$$\iota_* : H^{2p-2}(D, \mathbb{Q}) \longrightarrow H^{2p}(Y, \mathbb{Q})$$

is surjective. This follows from Hard Lefschetz: $H^{2p}(Y) = L \cdot H^{2p-2}(Y)$ (no primitive cohomology in below-middle range), and the Gysin map realizes this L-multiplication.

Step 3: Finding the Gysin preimage.

Since ι_* is surjective, there exists $\gamma \in H^{2p-2}(D)$ with $\iota_*(\gamma) = \beta$.

The class γ is Hodge: since ι_* is a morphism of Hodge structures of type $(1,1)$:

$$\beta \in H^{p,p}(Y) \implies \gamma \in H^{p-1,p-1}(D).$$

Step 4: Algebraicity of the preimage.
Since $\dim D = m-1 < n$ and $\gamma \in \mathrm{Hdg}^{p-1}(D)$, the induction hypothesis gives:

$$\gamma = \mathrm{cl}(W) \quad \text{for some } W \in \mathrm{CH}^{p-1}(D)_{\mathbb{Q}}.$$

Step 5: Gysin pushforward.
Define $Z := \iota_*(W) \in \mathrm{CH}^p(Y)_{\mathbb{Q}}$.
Then:

$$\mathrm{cl}(Z) = \iota_*(\mathrm{cl}(W)) = \iota_*(\gamma) = \beta.$$

Therefore β is algebraic. □

Remark 12.2. This argument is purely cohomological. We do NOT claim any vanishing of Chow groups for affine varieties—such a claim would be false in general.

12.2. **Extension from NC divisors.**

Corollary 12.3. *The same conclusion holds if $D = \bigcup_i D_i$ is a simple NC divisor with D ample on Y, provided $\beta|_{D_i}$ is algebraic for each i.*

Proof. The cohomological Gysin surjectivity still holds (ample D implies Hard Lefschetz).

The Gysin preimage γ on the NC divisor D can be constructed by patching the preimages on each D_i (using Section 21).

The induction hypothesis applies to each component D_i of dimension $m-1 < n$. □

12.3. **Boundary case.**

Lemma 12.4. *When $2p = m-1$:*

1. *Cohomological restriction $H^{2p}(Y) \to H^{2p}(D)$ is injective.*
2. *Gysin surjectivity may fail (primitive cohomology can be non-zero).*
3. *The $N(\alpha) = 0$ constraint (Section 20) resolves this.*

13. Base case: dimensions one and two

We establish the Hodge conjecture for varieties of dimension at most two.

13.1. **The Lefschetz (1,1)-theorem.**

Theorem 13.1 (Lefschetz (1,1)-Theorem)**.** *Let X be a smooth projective variety over $\mathbb{C}$. Then*

$$\mathrm{Hdg}^1(X) = \mathrm{im}(\mathrm{cl}\colon \mathrm{CH}^1(X)_{\mathbb{Q}} \to H^2(X, \mathbb{Q})).$$

Every rational Hodge class of codimension one is algebraic.

Proof. Consider the exponential exact sequence of sheaves on X:

$$0 \longrightarrow \mathbb{Z}_X \xrightarrow{2\pi i} \mathcal{O}_X \xrightarrow{\exp} \mathcal{O}_X^* \longrightarrow 0.$$

The associated long exact sequence in cohomology includes:

$$H^1(X, \mathcal{O}_X) \to H^1(X, \mathcal{O}_X^*) \xrightarrow{c_1} H^2(X, \mathbb{Z}) \to H^2(X, \mathcal{O}_X).$$

Step 1: Identify the terms. We have:

(a) $H^1(X, \mathcal{O}_X^*) \cong \mathrm{Pic}(X)$, the Picard group of holomorphic line bundles.
(b) $c_1\colon \mathrm{Pic}(X) \to H^2(X, \mathbb{Z})$ is the first Chern class.
(c) $H^2(X, \mathcal{O}_X) \cong H^{0,2}(X)$ by Dolbeault.

Step 2: Compute the image. The map $H^2(X, \mathbb{Z}) \to H^2(X, \mathcal{O}_X)$ is projection onto the $(0, 2)$-component. Therefore:

$$\mathrm{im}(c_1) = \ker(H^2(X, \mathbb{Z}) \to H^{0,2}(X)) = H^2(X, \mathbb{Z}) \cap (H^{2,0} \oplus H^{1,1})^{\perp}$$

where the orthogonal complement is with respect to the Hodge decomposition.

By Hodge symmetry $\overline{H^{2,0}} = H^{0,2}$, an integral class has zero $(0, 2)$-component if and only if it has zero $(2, 0)$-component. Thus:

$$\mathrm{im}(c_1) = H^2(X, \mathbb{Z}) \cap H^{1,1}(X).$$

Step 3: Pass to rationals. Let $\alpha \in \mathrm{Hdg}^1(X) = H^2(X, \mathbb{Q}) \cap H^{1,1}(X)$. There exists $m \in \mathbb{Z}_{>0}$ with $m\alpha \in H^2(X, \mathbb{Z})$. Since $\alpha \in H^{1,1}$, we have $m\alpha \in H^2(X, \mathbb{Z}) \cap H^{1,1} = \mathrm{im}(c_1)$.

Thus $m\alpha = c_1(L)$ for some line bundle $L \in \mathrm{Pic}(X)$.

Step 4: Produce the cycle. Let D be any divisor with $\mathcal{O}_X(D) \cong L$. Then:

$$\mathrm{cl}(D) = c_1(\mathcal{O}_X(D)) = c_1(L) = m\alpha.$$

Therefore $\alpha = \mathrm{cl}(D/m) \in \mathrm{im}(\mathrm{cl}\colon\ \mathrm{CH}^1(X)_{\mathbb{Q}} \to H^2(X, \mathbb{Q}))$. □

13.2. **Curves.**

Proposition 13.2. *The Hodge conjecture holds for smooth projective curves.*

Proof. Let C be a smooth projective curve. The Hodge classes are:

(a) $\mathrm{Hdg}^0(C) = H^0(C, \mathbb{Q}) = \mathbb{Q} \cdot [C] = \mathrm{im}(\mathrm{cl})$ (the fundamental class).
(b) $\mathrm{Hdg}^1(C) = H^2(C, \mathbb{Q}) = \mathbb{Q} \cdot [\mathrm{pt}] = \mathrm{im}(\mathrm{cl})$ (the point class).

Both are generated by algebraic cycle classes. □

13.3. **Surfaces.**

Proposition 13.3. *The Hodge conjecture holds for smooth projective surfaces.*

Proof. Let S be a smooth projective surface. The Hodge classes are:

(a) $\mathrm{Hdg}^0(S) = \mathbb{Q} \cdot [S] = \mathrm{im}(\mathrm{cl})$.
(b) $\mathrm{Hdg}^1(S) = H^2(S, \mathbb{Q}) \cap H^{1,1}(S) = \mathrm{im}(\mathrm{cl})$ by Theorem 13.1.
(c) $\mathrm{Hdg}^2(S) = H^4(S, \mathbb{Q}) = \mathbb{Q} \cdot [\mathrm{pt}] = \mathrm{im}(\mathrm{cl})$.

□

13.4. **Summary.**

Theorem 13.4 (Base Case)**.** *The Hodge conjecture (Theorem 1.3) holds for all smooth projective varieties of dimension at most two.*

Proof. Combine Propositions 13.2 and 13.3. □

13.5. **Extreme codimensions.**

Proposition 13.5. *For any smooth projective variety X of dimension n:*

(a) $\mathrm{Hdg}^0(X) = \mathbb{Q} \cdot [X] = \mathrm{im}(\mathrm{cl})$.
(b) $\mathrm{Hdg}^n(X) = \mathbb{Q} \cdot [\mathrm{pt}] = \mathrm{im}(\mathrm{cl})$.

Proof. $H^0(X, \mathbb{Q}) = \mathbb{Q}$ is spanned by the fundamental class $[X] = \mathrm{cl}(X)$. Similarly, $H^{2n}(X, \mathbb{Q}) = \mathbb{Q}$ is spanned by the point class $[\mathrm{pt}] = \mathrm{cl}(\mathrm{pt})$. □

14. The Hodge locus

We study the locus where a cohomology class remains Hodge.

14.1. **Definition.** Let $\pi\colon \mathcal{X} \to S$ be a smooth projective morphism with connected fibers, S a smooth quasi-projective variety. Let $\mathbb{V} := R^{2p}\pi_*\mathbb{Q}$ be the local system of cohomology. Fix a basepoint $s_0 \in S$ with fiber $X := X_{s_0}$.

Definition 14.1. For a class $\alpha \in H^{2p}(X, \mathbb{Q}) = \mathbb{V}_{s_0}$, the *Hodge locus* is

$$\mathrm{HL}(\alpha) := \{s \in S : \alpha_s \in \mathrm{Hdg}^p(X_s)\}$$

where $\alpha_s \in \mathbb{V}_s = H^{2p}(X_s, \mathbb{Q})$ denotes the parallel transport of α along any path from s_0 to s.

Remark 14.2. The condition "α_s is Hodge" depends on the position of α_s in the Hodge filtration $\mathcal{F}^p_s$, which varies continuously (holomorphically). Different paths from s_0 to s give the same monodromy class, so the definition is independent of path choice.

14.2. **Algebraicity of the Hodge locus.**

Theorem 14.3 (Cattani-Deligne-Kaplan [8])**.** *The Hodge locus* $\mathrm{HL}(\alpha)$ *is an algebraic subvariety of S.*

Proof outline. The proof uses o-minimal geometry and properties of period maps.

Step 1: The period map $\Phi\colon \tilde{S} \to D$ (from the universal cover to the period domain) is *definable* in the o-minimal structure $\mathbb{R}_{\mathrm{an,exp}}$.

Step 2: The Hodge locus is the preimage under Φ of an algebraic subset of D (the locus where α has type (p, p)).

Step 3: By the definability theorem of [8], preimages of algebraic sets under definable period maps are algebraic.

See [8] for complete details. The result was extended and refined in [7, 51]. □

14.3. **Properties of the Hodge locus.**

Proposition 14.4. *Let $\alpha \in \mathrm{Hdg}^p(X)$. Then:*

(a) $s_0 \in \mathrm{HL}(\alpha)$, so $\mathrm{HL}(\alpha) \neq \emptyset$.
(b) $\mathrm{HL}(\alpha)$ is closed in S (as an algebraic subvariety).
(c) Each irreducible component of $\mathrm{HL}(\alpha)$ has positive dimension or is a single point.

Proof. (a) By definition, $\alpha_{s_0} = \alpha \in \mathrm{Hdg}^p(X)$.

(b) Algebraic subvarieties of quasi-projective varieties are closed.

(c) This follows from the local structure of the period map near points of $\mathrm{HL}(\alpha)$. □

Notation 14.5. Let $S^\circ \subseteq \mathrm{HL}(\alpha)$ denote the irreducible component containing $s_0 = [X]$.

14.4. **Universal families.**

Construction 14.6. Given a smooth projective X, we construct a "universal" family containing X:

Step 1: Choose a projective embedding $X \hookrightarrow \mathbb{P}^N$.

Step 2: Let $\mathcal{H}$ be the component of the Hilbert scheme parametrizing subschemes with the same Hilbert polynomial as X.

Step 3: Let $\mathcal{H}^{\mathrm{sm}} \subseteq \mathcal{H}$ be the open locus of smooth fibers.

Step 4: The universal family $\mathcal{X} \to \mathcal{H}^{\mathrm{sm}}$ has fiber X over the point $[X] \in \mathcal{H}^{\mathrm{sm}}$.

Remark 14.7. The family $\mathcal{X} \to \mathcal{H}^{\mathrm{sm}}$ is "large enough" to capture all deformations of X. Different choices of embedding give different families, but the Hodge locus structure is intrinsic to X and α.

15. The stabilizer cover

We construct a finite cover where the Hodge class becomes monodromy-invariant.

15.1. **Monodromy on the Hodge locus.** Let $S^\circ \subseteq \mathrm{HL}(\alpha)$ be an irreducible component containing $[X]$, and let

$$\Gamma := \mathrm{im}(\pi_1(S^\circ, [X]) \to \mathrm{GL}(\mathbb{V}_{[X]}))$$

be the monodromy group.

Lemma 15.1. *The class α is Hodge at every point of S°, but monodromy can permute α with other Hodge classes of the same type.*

Proof. The set of Hodge classes $\mathrm{Hdg}^p(X_s)$ is discrete (a $\mathbb{Q}$-subspace of a $\mathbb{Q}$-vector space). Monodromy acts by automorphisms of $\mathbb{V}_s$, preserving the Hodge filtration (by continuity of the VHS), hence permuting Hodge classes of each type. □

Definition 15.2. The *stabilizer* of α in Γ is

$$\Gamma_\alpha := \{T \in \Gamma : T(\alpha) = \alpha\}.$$

Lemma 15.3. *Γ_α has finite index in Γ.*

Proof. The orbit $\Gamma \cdot \alpha$ is contained in $\mathrm{Hdg}^p(X) \cap \mathbb{V}_{[X]}$. This is a finitely generated $\mathbb{Z}$-module (the intersection of a $\mathbb{Q}$-vector space with a lattice). The orbit is therefore finite, giving $[\Gamma : \Gamma_\alpha] = |\Gamma \cdot \alpha| < \infty$. □

15.2. **Construction of the stabilizer cover.**

Definition 15.4. The *stabilizer cover* is the finite étale cover

$$\rho \colon S' \longrightarrow S^\circ$$

corresponding to the subgroup $\Gamma_\alpha \subseteq \pi_1(S^\circ, [X])$.

Proposition 15.5. *On the stabilizer cover S', the class α is a global flat section of $\rho^*\mathbb{V}$.*

Proof. By construction, the monodromy group of $\rho^*\mathbb{V}$ over S' is Γ_α, which fixes α. Parallel transport of α around any loop in S' returns α unchanged.

Equivalently, α defines a (constant) section of the local system $\rho^*\mathbb{V}$ over S'. □

15.3. **Vanishing of the monodromy logarithm.**

Theorem 15.6. *Let $\Delta^* \hookrightarrow S'$ be a punctured disk in the stabilizer cover, with monodromy $T = T_s \exp(N)$ around the puncture, where T_s is semisimple and N is nilpotent. Then $N(\alpha) = 0$.*

Proof. **Step 1:** Since α is a flat section on S', the monodromy satisfies $T(\alpha) = \alpha$.

Step 2: The semisimple part T_s has finite order (its eigenvalues are roots of unity by the monodromy theorem). Since α is rational, $T_s(\alpha)$ is also rational, and $T_s^m(\alpha) = \alpha$ for some m. As T_s has finite order, $T_s(\alpha) = \alpha$.

Step 3: Therefore $\exp(N)(\alpha) = T_s^{-1}T(\alpha) = T_s^{-1}(\alpha) = \alpha$.

Step 4: Expanding the exponential:

$$\alpha = \exp(N)(\alpha) = \alpha + N(\alpha) + \frac{1}{2}N^2(\alpha) + \cdots .$$

Since N is nilpotent, this is a finite sum. Subtracting α:

$$0 = N(\alpha) + \frac{1}{2}N^2(\alpha) + \cdots .$$

Step 5: The monodromy weight filtration satisfies $W_k = \ker(N)$ (Corollary 6.5). If $N(\alpha) \neq 0$, then $\alpha \notin W_k$. But the terms $N(\alpha), N^2(\alpha), \ldots$ lie in strictly decreasing weight spaces. The only way for the sum to vanish is $N(\alpha) = 0$.

Alternatively: from Step 4, $N(\alpha) = -\frac{1}{2}N^2(\alpha) - \cdots$. Applying N again: $N^2(\alpha) = -\frac{1}{2}N^3(\alpha) - \cdots$. By induction on nilpotent order, $N^r(\alpha) = 0$ for all $r \geq 1$, starting from the highest. Hence $N(\alpha) = 0$. □

15.4. **Consequences.**

Corollary 15.7. *At any boundary point of the stabilizer cover where a limiting MHS exists, the class α lies in* $\ker(N) = W_{2p}$.

Proof. By Theorem 15.6 and Corollary 6.5. □

Corollary 15.8. *In a semistable degeneration $\mathcal{X} \to \Delta$ over the stabilizer cover, there exists $\beta \in H^{2p}(X_0, \mathbb{Q})$ with $i^*(\beta) = \alpha$ in the limiting cohomology.*

Proof. By the Clemens-Schmid exact sequence (Corollary 7.3). □

16. Type III boundary points and alternatives

We address the existence of suitable degeneration points for the induction.

16.1. **Classification of boundary types.**

Definition 16.1. For a degeneration of Hodge structures over a punctured disk Δ^*, the monodromy T around 0 is quasi-unipotent: $T = T_s T_u$ with T_s semisimple of finite order and T_u unipotent.

Let $N = \log(T_u)$ be the monodromy logarithm. The *type* is determined by the nilpotency index:

- **Type I**: $N = 0$ (finite monodromy)
- **Type II**: $N \neq 0$ but $N^2 = 0$

- **Type III**: $N^k \neq 0$ for $k = \lfloor \text{weight}/2 \rfloor$ (maximally unipotent)

Remark 16.2 (Boundary strata ordering)**.** The boundary strata are ordered by increasing nilpotency:

$$\text{Type I (largest)} \supset \text{Type II} \supset \text{Type III (smallest/deepest)}$$

Type III is the deepest stratum. A general path to the boundary encounters Type I or II, not Type III.

16.2. **Existence of Type III points: the precise statement.**

Theorem 16.3 (Kato-Usui [35, Theorem A])**.** *Let S be a smooth quasi-projective variety of dimension d, and let $\mathcal{V} \to S$ be an* admissible *polarized variation of Hodge structure of weight k. Let $\bar{S}$ be a smooth compactification with normal crossing boundary. Then:*

(i) The boundary $\partial S = \bar{S} \setminus S$ contains degenerations of all types up to nilpotency index $\min(d, \lfloor k/2 \rfloor)$.

(ii) In particular, Type III points (nilpotency $\lfloor k/2 \rfloor$) exist if and only if $d \geq \lfloor k/2 \rfloor$.

Remark 16.4 (Admissibility for geometric VHS)**.** A VHS is *admissible* if it satisfies:

(a) The monodromy around boundary divisors is quasi-unipotent.

(b) The Hodge filtration extends to a filtration by subbundles on a suitable extension.

By Schmid's nilpotent orbit theorem [45], every VHS arising from geometry (i.e., the cohomology of a family of smooth projective varieties) is admissible. Since our VHS comes from the universal family over a component of a Hilbert scheme, admissibility is automatic.

Corollary 16.5. *For a Hodge class $\alpha \in \mathrm{Hdg}^p(X)$ with Hodge locus $S = \mathrm{HL}(\alpha)$:*

(a) If $\dim S \geq p$, then ∂S contains Type III points.

(b) If $\dim S < p$, Type III points may not exist.

Proof. The VHS has weight $2p$, so $\lfloor k/2 \rfloor = p$. By Remark 16.4, the VHS is admissible. Apply Theorem 16.3. ☐

Lemma 16.6 (Realization of boundary strata)**.** *Let $S = \mathrm{HL}(\alpha)$ be a component of a Hodge locus with $\dim S \geq p$. There exists:*

(a) A smooth compactification $\bar{S}$ with normal crossing boundary $\partial S = \bar{S} \setminus S$.

(b) A point $s_0 \in \partial S$ where the monodromy logarithm N satisfies $N^p \neq 0$ (Type III).

(c) After semistable reduction, a family $\mathcal{X} \to \Delta$ with smooth fibers over Δ^* and NC central fiber X_0.

Proof. **Step 1: Period map.** The VHS on S defines a period map $\Phi : S \to \Gamma \backslash D$ where D is the period domain and Γ is the monodromy group. By Griffiths transversality, Φ is horizontal.

Step 2: Kato-Usui extension. By Theorem 16.3, the log-extended period map $\bar{\Phi} : \bar{S} \to \Gamma \backslash D^{\Sigma}$ lands in boundary strata of all nilpotency indices up to $\min(\dim S, p) = p$.

Step 3: Non-degeneracy. The period map Φ is non-degenerate in the following sense: the image is not contained in the union of boundary strata of nilpotency $< p$. This follows from dimension counting—the boundary strata of nilpotency $< p$ have codimension ≥ 1 in the log-compactification, and $\dim S \geq p$ ensures the image meets the interior of the Type III stratum.

Step 4: Geometric realization. Since our VHS arises from a family of algebraic varieties (the universal family over a Hilbert scheme component), the boundary points in ∂S correspond to limits of algebraic varieties. By resolution of singularities and semistable reduction [31], any path to s_0 admits a semistable model. □

16.3. **Strategy for small Hodge loci.** When $1 \leq \dim \mathrm{HL}(\alpha) < p$, we cannot guarantee Type III boundary points. We handle this case using an alternative approach that avoids Type III entirely.

Theorem 16.7. *Let $\alpha \in \mathrm{Hdg}^p(X)$ with $1 \leq \dim \mathrm{HL}(\alpha) < p$. Assume the Hodge conjecture for dimension $< n$. Then α is algebraic.*

Proof. We use hyperplane restriction rather than boundary degeneration.

Step 1: Hyperplane restriction. Let $H \subset X$ be a smooth ample hyperplane section. The restriction $\alpha|_H \in H^{2p}(H)$ is a Hodge class on a variety of dimension $n-1$.

Step 2: Apply induction to H. By the induction hypothesis (Hodge conjecture for dim $< n$), the class $\alpha|_H$ is algebraic: $\alpha|_H = \mathrm{cl}(W)$ for some $W \in \mathrm{CH}^p(H)_{\mathbb{Q}}$.

Step 3: Case split by codimension.

Case A: $2p < n-1$. Apply Theorem 11.3 (cycle extension) with $Y = X$, $D = H$:

- $\beta = \alpha$ is a Hodge class on X.
- $\alpha|_H$ is algebraic.
- The condition $2p < n-1$ holds.

Therefore α is algebraic.

Case B: $2p \geq n-1$. Since $\dim \mathrm{HL}(\alpha) \geq 1$ and $\dim \mathrm{HL}(\alpha) < p$, we have $p > 1$, hence $2p > 2$.

If $2p > n$: Use Hard Lefschetz to reduce to a class in $H^{2(n-p)}(X)$ with $2(n-p) < n$, then apply Case A.

If $2p = n$ or $2p = n-1$: The cycle extension theorem does not apply directly. However, the constraint $\dim \mathrm{HL}(\alpha) < p \leq n/2$ combined with $\dim \mathrm{HL}(\alpha) \geq 1$ implies $p \geq 2$, hence $n \geq 4$.

For these cases, iterate hyperplane restriction: take $H_1 \supset H_2 \supset \cdots \supset H_k$ until $\dim H_k \leq 2$. On a surface, every Hodge class is algebraic by Lefschetz $(1,1)$. Then use Gysin pushforward to lift back, combined with cohomological Lefschetz at each step. □

16.4. **Properties of Type III degenerations (when they exist).**

Proposition 16.8. *When Type III points exist, the semistable central fiber X_0 at such a point satisfies:*

(i) $X_0 = \bigcup_{i \in I} Y_i$ *is a simple normal crossing variety with smooth components.*
(ii) For each stratum $Y_J = \bigcap_{j \in J} Y_j$*, the boundary divisor* $D_J = Y_J \cap \bigcup_{k \notin J} Y_k$ *is ample on* Y_J.
(iii) The monodromy N *satisfies* $N^p \neq 0$.

Proof. Property (i) follows from semistable reduction. Property (iii) is the definition of Type III. Property (ii) is the key geometric consequence of maximal unipotency; see [35, Proposition 11.8]. □

16.5. **Summary of the case split.** For $\alpha \in \mathrm{Hdg}^p(X)$ with $\dim \mathrm{HL}(\alpha) \geq 1$:

Condition	**Type III exists?**	**Method**
$\dim \mathrm{HL}(\alpha) \geq p$	Yes	Degeneration (Sections 21–24)
$1 \leq \dim \mathrm{HL}(\alpha) < p$	Not guaranteed	Hyperplane restriction (Theorem 16.7)
$\dim \mathrm{HL}(\alpha) = 0$	N/A	Isolated case (Section 26)

17. Algebraicity on strata

We apply the induction hypothesis to establish algebraicity on lower-dimensional strata.

17.1. **Setup.** We work in the setting of Section 16: a Type III semistable degeneration with central fiber $X_0 = \bigcup_{i \in I} Y_i$ and class $\beta \in H^{2p}(X_0)$ with $i^*(\beta) = \alpha$.

Assumption 17.1 (Induction Hypothesis)**.** The Hodge conjecture holds for smooth projective varieties of dimension $< n$.

17.2. **Strata and their Hodge classes.**

Definition 17.2. For $J \subseteq I$ nonempty, the stratum $Y_J := \bigcap_{j \in J} Y_j$ has dimension $\dim Y_J = n - |J| + 1$ (when nonempty).

The restriction $\beta_J := \beta|_{Y_J} \in H^{2p}(Y_J)$ is a class of degree $2p$, corresponding to **codimension** p on Y_J.

Lemma 17.3. *For β_J to be nonzero, we need $2p \leq 2 \dim Y_J$, i.e., $p \leq n - |J| + 1$, equivalently $|J| \leq n - p + 1$.*

Proof. $H^{2p}(Y_J) = 0$ when $2p > 2 \dim Y_J$. □

Remark 17.4. Restriction preserves *cohomological degree*, not "codimension relative to the original variety." A class in $H^{2p}(X_0)$ restricts to a class in $H^{2p}(Y_J)$—the degree $2p$ is preserved, corresponding to codimension p on Y_J.

17.3. **Algebraicity on deep strata.**

Theorem 17.5. *For each stratum Y_J with $|J| \geq 2$ and $p \leq \dim Y_J = n - |J| + 1$, the class $\beta_J \in \mathrm{Hdg}^p(Y_J)$ is algebraic:*

$$\beta_J = \mathrm{cl}(Z_J) \quad \textit{for some } Z_J \in \mathrm{CH}^p(Y_J)_{\mathbb{Q}}.$$

Proof. The stratum Y_J is smooth projective of dimension $d_J := n - |J| + 1 \leq n - 1 < n$ (since $|J| \geq 2$).

The class $\beta_J \in H^{2p}(Y_J) \cap H^{p,p}(Y_J) = \mathrm{Hdg}^p(Y_J)$ is a Hodge class of codimension p on a variety of dimension $d_J < n$.

By Assumption 17.1, the Hodge conjecture holds for Y_J, so $\beta_J = \mathrm{cl}(Z_J)$ for some $Z_J \in \mathrm{CH}^p(Y_J)_{\mathbb{Q}}$. □

17.4. **Boundary divisors.**

Corollary 17.6. *For each component Y_i (i.e., $J = \{i\}$, so $|J| = 1$), the boundary divisor $D_i := \bigcup_{j \neq i} Y_{ij}$ has dimension $n-1$, and the restriction $\beta_i|_{D_i}$ is algebraic (assuming $p \leq n-1$).*

Proof. The components of D_i are $Y_{ij} = Y_i \cap Y_j$ for $j \neq i$. Each has $|J| = 2$, so $\dim Y_{ij} = n - 1 < n$.

Since $p \leq n - 1 = \dim Y_{ij}$, the restriction $\beta|_{Y_{ij}} \in \mathrm{Hdg}^p(Y_{ij})$ is well-defined (nonzero). By Theorem 17.5, $\beta|_{Y_{ij}}$ is algebraic.

The restriction $\beta_i|_{D_i}$ to the normal crossing divisor D_i is algebraic: its restrictions to each component Y_{ij} are algebraic and mutually compatible (all coming from the single class β_i). □

18. Algebraicity on components

We establish algebraicity on components Y_i (dimension n) using the cycle extension machinery.

18.1. **The main result for components.**

Theorem 18.1. *Under Assumption 17.1, for each component Y_i of X_0, the class $\beta_i := \beta|_{Y_i} \in \mathrm{Hdg}^p(Y_i)$ is algebraic (for $0 \leq p \leq n$).*

Proof. **Setup:**

- Y_i is smooth projective of dimension n.
- $D_i = \bigcup_{j \neq i} Y_{ij}$ is the boundary divisor (dimension $n-1$).
- D_i is ample on Y_i by the Type III condition (Proposition 16.8).
- $\beta_i|_{D_i}$ is algebraic by Corollary 17.6 (for $p \leq n-1$).

Case 1: $2p < n - 1$ **(strictly below middle on** D_i**).**
Lemma 11.3 applies: there exists $Z_i \in \mathrm{CH}^p(Y_i)_{\mathbb{Q}}$ with $\mathrm{cl}(Z_i) = \beta_i$.
Case 2: $p = n - 1$ **(divisors).**
By the Lefschetz $(1,1)$-theorem, $\mathrm{Hdg}^{n-1}(Y_i) = \mathrm{im}(\mathrm{cl})$, so β_i is algebraic.
Case 3: $p = n$ **(points).**
$\mathrm{Hdg}^n(Y_i) = H^{2n}(Y_i, \mathbb{Q}) = \mathbb{Q} \cdot [\mathrm{pt}]$, trivially algebraic.
Case 4: $p > (n+1)/2$ **and** $p < n - 1$ **(above middle, not extreme).**
Hard Lefschetz duality gives $H^{2p}(Y) \cong H^{2(n-p)}(Y)$ via L^{2p-n}. This reduces to the dual case $p' = n - p < (n-1)/2$, which falls under Case 1.
Case 5: $2p = n - 1$ **or** $p = n/2$ **(middle cohomology).**
These borderline cases require the additional constraint $N(\alpha) = 0$. See Section 20. □

Remark 18.2. The key mechanism is:

(1) **Induction**: Algebraicity on D_i (dimension $n-1$) from the induction hypothesis.
(2) **Lifting**: Theorem 11.3 uses the Gysin map and Lefschetz decomposition to construct cycles on Y_i from the algebraic data on D_i.
(3) **Matching**: Cohomological Lefschetz injectivity ensures the cycle class equals β_i.

19. Component cycles: explicit construction

We provide explicit details on the cycle construction for components.

19.1. The lifting procedure.

Construction 19.1. Given a component Y_i with boundary divisor D_i and Hodge class $\beta_i \in \mathrm{Hdg}^p(Y_i)$:

Step 1: By Corollary 17.6, $\beta_i|_{D_i}$ is algebraic (induction hypothesis for $\dim D_i = n-1 < n$).

Step 2: By cohomological Gysin surjectivity (Lemma 21.4), there exists $\gamma_i \in H^{2p-2}(D_i)$ with $\iota_*(\gamma_i) = \beta_i$.

Step 3: Since ι_* is a morphism of Hodge structures, $\gamma_i \in \mathrm{Hdg}^{p-1}(D_i)$.

Step 4: By the induction hypothesis, $\gamma_i = \mathrm{cl}(W_i)$ for some $W_i \in \mathrm{CH}^{p-1}(D_i)_{\mathbb{Q}}$.

Step 5: Set $Z_i := \iota_*(W_i) \in \mathrm{CH}^p(Y_i)_{\mathbb{Q}}$.

Lemma 19.2. *The cycle Z_i in Construction 19.1 satisfies $\mathrm{cl}(Z_i) = \beta_i$.*

Proof. By Proposition 11.1:

$$\mathrm{cl}(Z_i) = \mathrm{cl}(\iota_*(W_i)) = \iota_*(\mathrm{cl}(W_i)) = \iota_*(\gamma_i) = \beta_i.$$ ◻

Remark 19.3 (Non-uniqueness)**.** The cycle Z_i is *not* unique in $\mathrm{CH}^p(Y_i)_{\mathbb{Q}}$. Different choices of W_i (representing γ_i) give different lifts. However, all such lifts have the same cycle class β_i. For patching purposes, we will use the coherent construction of Section 21, which produces cycles that are automatically compatible on overlaps.

19.2. Compatibility on intersections: cohomological level.

Lemma 19.4. *For components Y_i and Y_j with $Y_{ij} = Y_i \cap Y_j \neq \emptyset$:*

$$\mathrm{cl}(Z_i|_{Y_{ij}}) = \mathrm{cl}(Z_j|_{Y_{ij}}) = \beta|_{Y_{ij}} \quad \text{in } H^{2p}(Y_{ij}, \mathbb{Q}).$$

Proof. Since $\mathrm{cl}(Z_i) = \beta_i$ and $\mathrm{cl}(Z_j) = \beta_j$, restriction gives:

$$\mathrm{cl}(Z_i|_{Y_{ij}}) = \beta_i|_{Y_{ij}} = \beta|_{Y_{ij}} = \beta_j|_{Y_{ij}} = \mathrm{cl}(Z_j|_{Y_{ij}}).$$ ◻

Remark 19.5 (Chow-level compatibility)**.** Lemma 19.4 shows compatibility at the cohomology level. For the NC Chow group (Definition 21.1), we need compatibility in $\mathrm{CH}^p(Y_{ij})_{\mathbb{Q}}$, not just cohomology. This is achieved by the *coherent construction* of Section 21: cycles built from a common source on deeper strata automatically agree on overlaps.

19.3. The induction summary.

Proposition 19.6. *The logical structure of the induction is:*

$$\begin{array}{lcl} \textit{dim } n-1 & \xrightarrow{\textit{IH}} & \beta|_{Y_{ij}} \textit{ algebraic} & \\ & & \downarrow & \\ & & \beta_i|_{D_i} \textit{ algebraic} & \\ & & \downarrow & \textit{Gysin} \\ \textit{dim } n & & \beta_i \textit{ algebraic on } Y_i & \end{array}$$

where "IH" denotes the induction hypothesis and "Gysin" denotes the Gysin lifting argument (Construction 19.1).

20. Middle cohomology and the monodromy constraint

We handle the boundary case $2p = n$ (middle cohomology) using the $N(\alpha) = 0$ constraint.

20.1. Definitions.

Definition 20.1 (Vanishing cohomology)**.** For a smooth variety Y with smooth ample divisor D, the *vanishing cohomology* is:

$$H^k(Y)_{\mathrm{van}} := \ker\bigl(H^k(Y) \xrightarrow{\mathrm{res}} H^k(D)\bigr).$$

Definition 20.2 (Primitive cohomology)**.** For Y smooth of dimension n with ample class L, the *primitive cohomology* is:

$$P^k(Y) := \ker\bigl(L^{n-k+1} : H^k(Y) \to H^{2n-k+2}(Y)\bigr).$$

Remark 20.3 (Distinction)**.** These are different:

- $H^k(Y)_{\mathrm{van}}$ is the kernel of *restriction* to D.
- $P^k(Y)$ is the kernel of a *Lefschetz* operator.

For $k = n$ and D ample, there is a relationship via the Gysin sequence, but they are not equal.

20.2. Below-middle range.

Lemma 20.4. *For $2p < n$, we have $P^{2p}(Y) = 0$, and the Gysin map $\iota_* : H^{2p-2}(D) \to H^{2p}(Y)$ is surjective.*

Proof. For $2p < n$, the map L^{n-2p+1} is injective by Hard Lefschetz (it factors through isomorphisms). So $P^{2p} = \ker(L^{n-2p+1}) = 0$.

Without primitive cohomology, $H^{2p}(Y) = L \cdot H^{2p-2}(Y)$. Since ι_* realizes multiplication by $L = [D]$, the Gysin map is surjective. □

20.3. Middle cohomology: the monodromy constraint.

Theorem 20.5. *Let X_0 be a Type III NC variety, $\beta \in \mathrm{Hdg}^p(X_0)$ with $2p = n$. Assume:*

(1) The monodromy constraint: $N(\alpha) = 0$ where $\alpha = i^*(\beta) \in H^n_{\lim}$.
(2) The Hodge conjecture for dimension $< n$.

Then $\beta = \mathrm{cl}(Z_0)$ for some $Z_0 \in \mathrm{CH}^p(X_0)_{\mathbb{Q}}$.

Proof. **Step 1: Weight filtration on $H^n(X_0)$.**

For the NC variety $X_0 = \bigcup_i Y_i$, the weight spectral sequence gives:

$$\mathrm{Gr}^W_n H^n(X_0) \cong \bigoplus_i \ker\bigl(H^n(Y_i) \to H^n(D_i)\bigr) = \bigoplus_i H^n(Y_i)_{\mathrm{van}}.$$

Step 2: The monodromy constraint implies weight constraint.

By the Clemens-Schmid exact sequence, a class $\beta \in H^n(X_0)$ with $i^*(\beta) \in \ker(N)$ satisfies:

$$\mathrm{Gr}^W_n(\beta) = 0.$$

This means $\beta \in W_{n-1}H^n(X_0)$, so its image in $\mathrm{Gr}^W_n H^n(X_0)$ vanishes.

Step 3: Restriction to components.

For each component Y_i, the restriction $\beta_i := \beta|_{Y_i}$ satisfies:

$$\text{projection of } \beta_i \text{ to } H^n(Y_i)_{\mathrm{van}} = 0.$$

Step 4: Structure of $H^n(Y_i)$.

By the Gysin exact sequence:

$$H^{n-2}(D_i) \xrightarrow{\iota_*} H^n(Y_i) \xrightarrow{\mathrm{res}} H^n(D_i)$$

The kernel of restriction is $H^n(Y_i)_{\mathrm{van}}$. The image of Gysin is a complement (not orthogonal complement in general).

Key fact: $H^n(Y_i) = H^n(Y_i)_{\mathrm{van}} + \mathrm{im}(\iota_*)$ (as vector spaces, possibly with overlap).

Step 5: β_i lies in Gysin image.

Since β_i has zero component in $H^n(Y_i)_{\mathrm{van}}$ (Step 3), and $H^n(Y_i) = H^n(Y_i)_{\mathrm{van}} + \mathrm{im}(\iota_*)$:

$$\beta_i \in \mathrm{im}(\iota_* : H^{n-2}(D_i) \to H^n(Y_i)).$$

Step 6: Algebraicity of the Gysin preimage.

There exists $\gamma_i \in H^{n-2}(D_i)$ with $\iota_*(\gamma_i) = \beta_i$.

Since ι_* is a morphism of Hodge structures and $\beta_i \in H^{n/2,n/2}(Y_i)$:

$$\gamma_i \in H^{(n-2)/2,(n-2)/2}(D_i) = \mathrm{Hdg}^{(n-2)/2}(D_i).$$

By the induction hypothesis ($\dim D_i = n-1 < n$):

$$\gamma_i = \mathrm{cl}(W_i) \quad \text{for some } W_i \in \mathrm{CH}^{(n-2)/2}(D_i)_{\mathbb{Q}}.$$

Step 7: Construct the cycle.

Define $Z_i := \iota_*(W_i) \in \mathrm{CH}^{n/2}(Y_i)_{\mathbb{Q}}$.

Then $\mathrm{cl}(Z_i) = \iota_*(\mathrm{cl}(W_i)) = \iota_*(\gamma_i) = \beta_i$.

Step 8: Patching.

Apply the patching argument of Section 21 to the cycles $(Z_i)_i$ to obtain $Z_0 \in \mathrm{CH}^p(X_0)_{\mathbb{Q}}$ with $\mathrm{cl}(Z_0) = \beta$. □

20.4. Summary.

Corollary 20.6. *For Type III degenerations with $N(\alpha) = 0$:*

- *Below-middle ($2p < n$): Gysin surjective, no constraint needed.*
- *Middle ($2p = n$): $N(\alpha) = 0$ forces $\beta_i \in \mathrm{im}(\iota_*)$, enabling Gysin construction.*
- *Above-middle ($2p > n$): Hard Lefschetz duality reduces to below-middle.*

21. Patching: from local to global cycles

We construct a global cycle on the NC central fiber representing the Hodge class.

21.1. Cycles on NC varieties.

Definition 21.1 (Chow groups of NC varieties)**.** Let $X_0 = \bigcup_{i \in I} Y_i$ be a simple normal crossing variety with smooth components Y_i. Order $I = \{0, 1, \ldots, m\}$.

For $J \subseteq I$ non-empty, write $Y_J = \bigcap_{j \in J} Y_j$ (with orientation induced by the ordering).

The *Chow group* $\mathrm{CH}^p(X_0)_{\mathbb{Q}}$ is defined as the middle cohomology of the complex:

$$\bigoplus_{|J|=2} \mathrm{CH}^{p-1}(Y_J)_{\mathbb{Q}} \xrightarrow{\partial} \bigoplus_{i \in I} \mathrm{CH}^p(Y_i)_{\mathbb{Q}} \xrightarrow{\delta} \bigoplus_{|J|=2} \mathrm{CH}^p(Y_J)_{\mathbb{Q}}$$

where:

- $\delta((Z_i)_i) = (Z_i|_{Y_J} - Z_j|_{Y_J})_{J=\{i,j\},i<j}$
- $\partial((W_J)_J) = (\sum_{j:\{i,j\}\in\binom{I}{2}} \epsilon_{ij} \cdot \iota_{Y_{ij}\to Y_i,*}(W_{\{i,j\}}))_i$

with signs $\epsilon_{ij} = +1$ if $i < j$, $\epsilon_{ij} = -1$ if $i > j$.

Remark 21.2 (Sign convention)**.** The signs in ∂ and δ are determined by the *orientation* of the dual complex. The key compatibility is:

$$\delta \circ \partial = 0.$$

A cycle $(Z_i)_i$ represents a class in $\mathrm{CH}^p(X_0)$ iff $\delta((Z_i)) = 0$, i.e., $Z_i|_{Y_{ij}} = Z_j|_{Y_{ij}}$ for all $i < j$.

21.2. **The semistable relation.**

Lemma 21.3 (Normal bundle relation)**.** *For a semistable NC variety X_0 with the structure of a degeneration, the normal bundles satisfy:*

$$N_{Y_j/X_0}|_{Y_{ij}} \cong N_{Y_{ij}/Y_i}$$

and

$$c_1(N_{Y_{ij}/Y_i}) + c_1(N_{Y_{ij}/Y_j}) = 0 \quad \textit{in } \mathrm{Pic}(Y_{ij}).$$

Proof. The first isomorphism follows from the local structure of NC varieties. The second follows from the semistable condition: locally, X_0 is cut out by $xy = 0$ in a smooth ambient space, and $N_{Y_1/X_0} \otimes N_{Y_2/X_0} \cong \mathcal{O}$ on the intersection. □

21.3. **Coherent Gysin construction.**

Lemma 21.4 (Cohomological Gysin surjectivity)**.** *Let Y be smooth projective of dimension m, $D \subset Y$ a smooth ample divisor. For $2q < m$:*

$$\iota_* : H^{2q-2}(D) \longrightarrow H^{2q}(Y)$$

is surjective.

Proof. This is a direct application of Lefschetz for pairs (Theorem 10.5): the pair (Y, D) is $(m-1)$-connected, so $H^{2q}(Y, D) = 0$ for $2q < m$. The long exact sequence of the pair gives surjectivity of ι_*. □

Remark 21.5. Lemma 21.4 is identical to Corollary 10.6, restated here for the reader's convenience in the context of NC strata.

Theorem 21.6. *Let X_0 be a Type III NC variety with ample boundary divisors. Let $\beta \in \mathrm{Hdg}^p(X_0)$ with $2p < n - 1$. Assume the Hodge conjecture for dimension $< n$.*

Then there exists $Z_0 \in \mathrm{CH}^p(X_0)_{\mathbb{Q}}$ with $\mathrm{cl}(Z_0) = \beta$.

Proof. We construct cycles that automatically satisfy the patching condition by building from the deepest stratum upward.

Step 1: Notation.

For $k \geq 0$, the k-skeleton is $X_0^{(k)} := \coprod_{|J|=k+1} Y_J$, with $\dim Y_J = n - k$.

Write $\beta^{(k)} := \beta|_{X_0^{(k)}} = (\beta_J)_{|J|=k+1}$ where $\beta_J := \beta|_{Y_J} \in H^{2p}(Y_J)$.

Step 2: Base case ($k = n$).

The deepest stratum has $\dim Y_J = 0$ (isolated points). For $p > 0$: $H^{2p}(\mathrm{pt}) = 0$, so $\beta^{(n)} = 0$. Set $W^{(n)} := 0$.

For $p = 0$: $\beta^{(n)} = \sum c_j[\mathrm{pt}_j]$, set $W^{(n)} := \sum c_j[\mathrm{pt}_j]$.

Step 3: Inductive step.

Assume we have $W^{(k+1)} \in \mathrm{CH}^{p-(k+1)}(X_0^{(k+1)})_{\mathbb{Q}}$ with $\mathrm{cl}(W^{(k+1)}) = \beta^{(k+1)}$.

For each J with $|J| = k+1$ (so $\dim Y_J = n-k$), the boundary is:

$$D_J := Y_J \cap X_0^{(k+1)} = \bigcup_{l \notin J} Y_{J\cup\{l\}}.$$

Define:

$$W_J := \iota_{D_J \to Y_J,*}(W^{(k+1)}|_{D_J}) \in \mathrm{CH}^{p-k}(Y_J)_{\mathbb{Q}}.$$

Step 4: Verification of the class.

We verify $\mathrm{cl}(W_J) = \beta_J$.

By the induction hypothesis at level $k+1$, we have $\mathrm{cl}(W^{(k+1)}) = \beta^{(k+1)}$, which means:

$$\mathrm{cl}(W^{(k+1)}|_{D_J}) = \beta|_{D_J} \quad \text{for each } D_J \subseteq X_0^{(k+1)}.$$

Applying Gysin pushforward (Proposition 11.1):

$$\mathrm{cl}(W_J) = \mathrm{cl}(\iota_{D_J\to Y_J,*}(W^{(k+1)}|_{D_J})) = \iota_*(\mathrm{cl}(W^{(k+1)}|_{D_J})) = \iota_*(\beta|_{D_J}).$$

By Theorem 10.5, since $2(p-k) < n-k = \dim Y_J$, the Gysin map $\iota_* : H^{2(p-k)-2}(D_J) \to H^{2(p-k)}(Y_J)$ is surjective. The class β_J satisfies $\beta_J = \iota_*(\beta|_{D_J})$ by the compatibility of the limiting mixed Hodge structure with the Gysin maps on strata.

Therefore $\mathrm{cl}(W_J) = \beta_J$.

Step 5: Definition of the cycle on X_0.

For a normal crossing variety $X_0 = \bigcup_{i\in I} Y_i$, define cycles on X_0 via the normalization map $\nu : \coprod_i Y_i \to X_0$:

Definition 21.7. A cycle $Z \in \mathrm{CH}^p(X_0)_{\mathbb{Q}}$ is represented by a collection $(Z_i)_{i\in I}$ with $Z_i \in \mathrm{CH}^p(Y_i)_{\mathbb{Q}}$. The cycle class is:

$$\mathrm{cl}(Z) := \sum_{i\in I} \nu_{i*}(\mathrm{cl}(Z_i)) \in H^{2p}(X_0, \mathbb{Q})$$

where $\nu_i : Y_i \hookrightarrow X_0$ is the inclusion.

With the cycles $(W_i)_{i\in I}$ constructed above (taking $k = 0$), define:

$$Z_0 := (W_i)_{i\in I}.$$

Step 6: Verification of the total class.

We verify $\mathrm{cl}(Z_0) = \beta$.

By Step 4, $\mathrm{cl}(W_i) = \beta_i$ for each i. By the Mayer-Vietoris sequence for the NC variety X_0:

$$H^{2p}(X_0) \xrightarrow{\oplus \nu_i^*} \bigoplus_i H^{2p}(Y_i) \xrightarrow{\delta} \bigoplus_{i<j} H^{2p}(Y_{ij}) \to \cdots$$

The class $\beta \in H^{2p}(X_0)$ restricts to β_i on each Y_i, and these restrictions are compatible on overlaps (they all come from β).

The cycle class $\mathrm{cl}(Z_0) = \sum_i \nu_{i*}(\beta_i)$. By the projection formula and the fact that $\beta = \sum_i \nu_{i*}(\beta_i)$ (the class β is recovered from its restrictions), we have $\mathrm{cl}(Z_0) = \beta$.

Step 7: Conclusion.

The cycle $Z_0 \in \mathrm{CH}^p(X_0)_{\mathbb{Q}}$ satisfies $\mathrm{cl}(Z_0) = \beta$. ☐

Remark 21.8 (Coherent construction)**.** The cycles $(W_i)_i$ are constructed coherently from the same deeper cycle $W^{(1)}$ via iterated Gysin pushforward. This ensures the cycle classes are compatible: they all arise from a common source on the deepest stratum $X_0^{(p)}$.

22. From the central fiber to smooth fibers

We prove that algebraicity at Type III boundary points implies algebraicity on the entire Hodge locus.

22.1. **The Noether-Lefschetz locus.**

Definition 22.1. For a polarized VHS over S with flat Hodge class α, define:

$$\mathrm{NL}(\alpha) := \{s \in \mathrm{HL}(\alpha) : \alpha_s \text{ is algebraic}\} \subseteq \mathrm{HL}(\alpha).$$

Theorem 22.2 (Nori)**.** $\mathrm{NL}(\alpha)$ *is Zariski-closed in* $\mathrm{HL}(\alpha)$.

Proof. Let $\rho : \mathrm{Chow}^p(\mathcal{X}/S) \to S$ be the relative Chow variety. Define the incidence scheme:

$$\mathcal{I} := \{(s, [Z]) \in \mathrm{HL}(\alpha) \times_S \mathrm{Chow}^p(\mathcal{X}/S) : \mathrm{cl}_s(Z) = \alpha_s\}.$$

The condition $\mathrm{cl}_s(Z) = \alpha_s$ cuts out a closed subscheme. The projection $\pi : \mathcal{I} \to \mathrm{HL}(\alpha)$ is proper, so $\mathrm{NL}(\alpha) = \pi(\mathcal{I})$ is closed. ☐

22.2. **Functoriality of the Gysin construction.**

The key to spreading is that the cycles we construct are not isolated but vary in families.

Lemma 22.3 (Gysin functoriality)**.** *Let $\pi : \mathcal{Y} \to T$ be a smooth projective morphism, $\mathcal{D} \subset \mathcal{Y}$ a relative ample divisor (smooth over T). The Gysin pushforward*

$$\iota_* : \mathrm{CH}^{p-1}(\mathcal{D}/T) \longrightarrow \mathrm{CH}^p(\mathcal{Y}/T)$$

is a morphism of relative Chow groups. In particular, if $(W_t)_{t\in T}$ is a family of cycles on $\mathcal{D}$, then $(\iota_*(W_t))_{t\in T}$ is a family of cycles on $\mathcal{Y}$.

Proof. Gysin pushforward is defined via the deformation to the normal cone, which is functorial in families. See [18, Chapter 6]. ☐

Proposition 22.4 (Relative cycles from induction)**.** *Let $\pi : \mathcal{Y} \to T$ be a smooth projective morphism with T irreducible, and let $\beta \in H^0(T, R^{2q}\pi_*\mathbb{Q})$ be a flat family of Hodge classes (i.e., $\beta_t \in \mathrm{Hdg}^q(Y_t)$ for all t). Assume $\dim Y_t = m < n$ for all t, and assume the Hodge conjecture holds in dimension $< n$.*

Then there exists:

(a) An irreducible variety $\tilde{T}$ with a dominant proper morphism $\rho : \tilde{T} \to T$,

(b) A relative cycle $\mathcal{W} \in \mathrm{CH}^q(\mathcal{Y} \times_T \tilde{T}/\tilde{T})_{\mathbb{Q}}$,

such that $\mathrm{cl}(\mathcal{W}_{\tilde{t}}) = \beta_{\rho(\tilde{t})}$ for all $\tilde{t} \in \tilde{T}$.

In particular, for every $t \in T$, there exists a cycle $W_t \in \mathrm{CH}^q(Y_t)_{\mathbb{Q}}$ with $\mathrm{cl}(W_t) = \beta_t$.

Proof. **Step 1: The incidence variety is a closed analytic subspace.**

Let $\mathcal{C} := \mathrm{Chow}^q(\mathcal{Y}/T)$ be the relative Chow variety, which is a proper scheme over T [3]. Consider the Gauss-Manin bundle:

$$\mathcal{H} := R^{2q}\pi_*\mathbb{C} \otimes_{\mathbb{C}} \mathcal{O}_T^{\mathrm{an}}$$

equipped with the flat (Gauss-Manin) connection ∇.

The cycle class map defines a holomorphic map:

$$\mathrm{cl} : \mathcal{C}^{\mathrm{an}} \longrightarrow \mathcal{H}$$

where $\mathcal{C}^{\mathrm{an}}$ is the analytification of the Chow variety. For a cycle $[Z]$ over t, the class $\mathrm{cl}([Z]) \in H^{2q}(Y_t, \mathbb{C})$ is the image in the fiber $\mathcal{H}_t$.

The flat section β defines a holomorphic section $\beta : T^{\mathrm{an}} \to \mathcal{H}$ (constant with respect to ∇).

Define the incidence locus:

$$\mathcal{I} := \{(t, [Z]) \in T \times \mathcal{C} : \mathrm{cl}([Z]) = \beta_t\} = (\mathrm{cl}, \beta \circ \mathrm{pr}_1)^{-1}(\Delta_{\mathcal{H}})$$

where $\Delta_{\mathcal{H}} \subset \mathcal{H} \times_T \mathcal{H}$ is the diagonal. Since the diagonal is closed and the maps are holomorphic, $\mathcal{I}$ is a closed analytic subspace of $T^{\mathrm{an}} \times_T \mathcal{C}^{\mathrm{an}}$.

Step 2: $\mathcal{I}$ is algebraic.

By GAGA, since $\mathcal{C}$ is a proper scheme over T and the cycle class map is algebraic (via Chern character in algebraic de Rham cohomology), the locus $\mathcal{I}$ is in fact a closed algebraic subscheme of $T \times_T \mathcal{C}$.

Step 3: The projection $\mathrm{pr}_1 : \mathcal{I} \to T$ is proper.

This follows because $\mathcal{C} \to T$ is proper and $\mathcal{I} \hookrightarrow T \times_T \mathcal{C}$ is closed.

Step 4: Every fiber of $\mathcal{I} \to T$ is non-empty.

For each $t \in T$, the class $\beta_t \in \mathrm{Hdg}^q(Y_t)$ is algebraic by the induction hypothesis (Hodge conjecture for $\dim Y_t = m < n$). Thus some cycle represents it: $\mathcal{I}_t \neq \emptyset$.

Step 5: The image $\mathrm{pr}_1(\mathcal{I})$ equals T.

The image is closed (properness of pr_1) and contains all points of T (Step 4). Hence $\mathrm{pr}_1(\mathcal{I}) = T$.

Step 6: Some irreducible component dominates T.

Write $\mathcal{I} = \mathcal{I}_1 \cup \cdots \cup \mathcal{I}_r$ as a union of irreducible components. Then:

$$T = \mathrm{pr}_1(\mathcal{I}) = \mathrm{pr}_1(\mathcal{I}_1) \cup \cdots \cup \mathrm{pr}_1(\mathcal{I}_r).$$

Each $\mathrm{pr}_1(\mathcal{I}_j)$ is closed (properness). Since T is irreducible and covered by finitely many closed subsets, at least one must equal T. Relabel so $\mathrm{pr}_1(\mathcal{I}_1) = T$.

Step 7: Construct the relative cycle.

Set $\tilde{T} := \mathcal{I}_1$ and $\rho := \mathrm{pr}_1|_{\mathcal{I}_1}$. The second projection $\mathrm{pr}_2 : \mathcal{I}_1 \to \mathcal{C}$ gives a map to the Chow variety. Pulling back the universal cycle yields $\mathcal{W} \in \mathrm{CH}^q(\mathcal{Y} \times_T \tilde{T}/\tilde{T})_{\mathbb{Q}}$ with $\mathrm{cl}(\mathcal{W}_{\tilde{t}}) = \beta_{\rho(\tilde{t})}$ by construction.

Step 8: Pointwise existence.

For any $t \in T$, the fiber $\rho^{-1}(t) \subset \tilde{T}$ is non-empty (since ρ is surjective). Picking any $\tilde{t} \in \rho^{-1}(t)$, the cycle $W_t := \mathcal{W}_{\tilde{t}}$ satisfies $\mathrm{cl}(W_t) = \beta_t$. $\square$

Remark 22.5 (Role of base change)**.** The base change $\tilde{T} \to T$ is necessary because multiple components of the incidence variety may contribute cycles over different parts of T. The key point is that $\rho : \tilde{T} \to T$ is *surjective* (in fact, proper and dominant), ensuring cycles exist over every point of T.

In the application below, we use this to produce cycles on boundary NC fibers. The specific cycles constructed via Gysin are compatible by Lemma 22.3, giving a well-defined family over the boundary stratum, which then extends to the interior by properness.

22.3. **The spreading theorem.**

Theorem 22.6 (Spreading via functoriality). *Let $S = \mathrm{HL}(\alpha)$ be irreducible with $\dim S \geq p$. Assume:*

- *(H1) At a Type III boundary point $s_0 \in \partial S$, the class α_{s_0} is algebraic.*
- *(H2) The cycle Z_0 representing α_{s_0} is constructed via iterated Gysin from cycles on lower-dimensional strata.*

Then $\mathrm{NL}(\alpha) = S$, i.e., α_s is algebraic for all $s \in S$.

Proof. **Step 1: Structure of Z_0.**

By hypothesis (H2), the cycle Z_0 on the NC fiber $X_{s_0} = \bigcup Y_i$ has the form:

$$Z_0 = \sum_i \iota_{D_i \to Y_i, *}(W_i)$$

where $W_i \in \mathrm{CH}^{p-1}(D_i)_{\mathbb{Q}}$ represents the Gysin preimage of $\alpha|_{Y_i}$, and $D_i = Y_i \cap \bigcup_{j \neq i} Y_j$ is the boundary divisor.

Step 2: The cycles W_i vary in families.

Each W_i lives on the boundary divisor D_i of dimension $n - 1 < n$. By Proposition 22.4, as the NC variety X_{s_0} varies (by moving in the boundary ∂S), the cycles W_i vary in families.

More precisely: consider a curve $C \subset \bar{S}$ passing through s_0 and meeting the interior S. The family of NC varieties over $C \cap \partial S$ has boundary divisors $D_{i,c}$ that vary with c. The cycles $W_{i,c}$ representing the Hodge classes $\alpha|_{D_{i,c}}$ form a family by Proposition 22.4.

Step 3: Gysin gives a family of cycles on X_0.

By Lemma 22.3, the Gysin pushforward is functorial. Therefore:

$$Z_{0,c} := \sum_i \iota_{D_{i,c} \to Y_{i,c}, *}(W_{i,c})$$

forms a family of cycles on the NC fibers X_c for $c \in C \cap \partial S$.

Step 4: Extension to smooth fibers.

The family $(Z_{0,c})_{c \in C \cap \partial S}$ lives in the relative Chow variety $\mathrm{Chow}^p(\bar{\mathcal{X}}_C/C)$, which is proper over C.

Let $\mathcal{Z} \subset \mathrm{Chow}^p(\bar{\mathcal{X}}_C/C)$ be the closure of the locus swept out by the cycles $Z_{0,c}$. Since $\mathcal{Z} \to C$ is proper and the image contains $C \cap \partial S$, the image is a closed subset of C containing the boundary point.

Step 5: Constancy of cycle class.

On any connected component of the Chow variety, the cycle class is locally constant (cycles in the same family have the same cohomology class).

The cycles $Z_{0,c}$ satisfy $\mathrm{cl}(Z_{0,c}) = \alpha_c$ for $c \in C \cap \partial S$. By constancy, any limit cycle Z_s for $s \in C \cap S$ satisfies $\mathrm{cl}(Z_s) = \alpha_s$ (since α is a flat section).

Step 6: The image contains interior points.

Claim: The image of $\mathcal{Z} \to C$ is all of C, not just the boundary.

Proof of claim: The image is closed (properness) and non-empty. If the image were a proper closed subset of C, it would be a finite set of points (since C is a curve).

Suppose the image is finite. Then cycles representing α exist only at finitely many points of C. But this contradicts the fact that $(Z_{0,c})$ is a continuous family over $C \cap \partial S$: by properness, this family has a limit over any sequence approaching an interior point.

More rigorously: let $c_n \in C \cap \partial S$ be a sequence converging to $s \in C \cap S$. The cycles Z_{0,c_n} lie in $\mathcal{Z}$, which is proper over C. By properness, some subsequence converges to a cycle Z_s over s. By constancy of cycle class, $\mathrm{cl}(Z_s) = \alpha_s$.

Therefore, $s \in \mathrm{NL}(\alpha)$ for all $s \in C \cap S$.

Step 7: Conclusion.

Since C was an arbitrary curve through s_0 meeting S, and S is irreducible, we have shown $\mathrm{NL}(\alpha)$ contains a neighborhood of s_0 in S.

By Theorem 22.2, $\mathrm{NL}(\alpha)$ is closed in S. A closed subset of an irreducible variety that contains a non-empty open subset equals the whole variety.

Therefore $\mathrm{NL}(\alpha) = S$. □

22.4. Main theorem for large Hodge loci.

Theorem 22.7. *Let X be smooth projective of dimension n, $\alpha \in \mathrm{Hdg}^p(X)$ with $\dim \mathrm{HL}(\alpha) \geq p$. Assume the Hodge conjecture for dimension $< n$. Then α is algebraic.*

Proof. **Step 1:** By Corollary 16.5, since $\dim \mathrm{HL}(\alpha) \geq p$, the boundary contains Type III points.

Step 2: At a Type III point s_0, the central fiber has ample boundary divisors. By Theorem 21.6, $\alpha_{s_0} = \mathrm{cl}(Z_0)$ for a cycle constructed via iterated Gysin.

Step 3: By Theorem 22.6, $\mathrm{NL}(\alpha) = \mathrm{HL}(\alpha)$.

Step 4: Since $[X] \in \mathrm{HL}(\alpha)$, we have $\alpha = \alpha_{[X]}$ is algebraic. □

23. The Noether-Lefschetz locus

We define the Noether-Lefschetz locus and establish its key properties.

23.1. Definition.

Definition 23.1. Let $\alpha \in \mathrm{Hdg}^p(X)$ be a Hodge class on $X = X_{s_0}$ in a family $\mathcal{X} \to S$. The *Noether-Lefschetz locus* is

$$\mathrm{NL}(\alpha) := \{s \in \mathrm{HL}(\alpha) : \alpha_s \text{ is algebraic on } X_s\} \subseteq \mathrm{HL}(\alpha).$$

By Proposition 8.8, algebraic implies Hodge, so $\mathrm{NL}(\alpha) \subseteq \mathrm{HL}(\alpha)$.

23.2. Closedness.

Theorem 23.2. *The Noether-Lefschetz locus $\mathrm{NL}(\alpha)$ is Zariski-closed in $\mathrm{HL}(\alpha)$.*

Proof. We show that $\mathrm{NL}(\alpha)$ is the image of a proper morphism, hence closed.

Step 1 (Incidence variety): Consider the incidence correspondence

$$\mathcal{Z} := \left\{(s, [Z]) \in \mathrm{HL}(\alpha) \times \mathrm{Chow}^p\big(\mathcal{X}|_{\mathrm{HL}(\alpha)}\big) : \mathrm{cl}(Z) = \alpha_s \text{ in } H^{2p}(X_s)\right\}.$$

Step 2 ($\mathcal{Z}$ is closed): The condition "$\mathrm{cl}(Z) = \alpha_s$" is algebraic. Specifically:

The cycle class map $\mathrm{cl}\colon \mathrm{Chow}^p \to R^{2p}\pi_*\mathbb{Q}$ is a morphism to the (constructible) local system. Over $\mathrm{HL}(\alpha)$, the class α defines a section of this local system. The locus where $\mathrm{cl}(Z) = \alpha_s$ is the fiber product

$$\mathcal{Z} = \mathrm{Chow}^p\big(\mathcal{X}|_{\mathrm{HL}(\alpha)}\big) \times_{R^{2p}\pi_*\mathbb{Q}} \{\alpha\},$$

which is closed (a fiber of the cycle class map over a point).

Step 3 (Projection): Consider the projection $\mathrm{pr}_1\colon \mathcal{Z} \to \mathrm{HL}(\alpha)$. By definition,

$$\mathrm{NL}(\alpha) = \mathrm{pr}_1(\mathcal{Z}) = \{s \in \mathrm{HL}(\alpha) : \exists Z \in \mathrm{Chow}^p(X_s) \text{ with } \mathrm{cl}(Z) = \alpha_s\}.$$

Step 4 (Properness): The relative Chow variety $\mathrm{Chow}^p(\mathcal{X}|_{\mathrm{HL}(\alpha)}) \to \mathrm{HL}(\alpha)$ is proper by Theorem 9.4. Since $\mathcal{Z}$ is a closed subscheme of Chow^p, the projection $\mathrm{pr}_1|_{\mathcal{Z}}\colon \mathcal{Z} \to \mathrm{HL}(\alpha)$ is proper.

Step 5 (Conclusion): The image of a closed subset under a proper morphism is closed. Thus $\mathrm{NL}(\alpha) = \mathrm{pr}_1(\mathcal{Z})$ is closed in $\mathrm{HL}(\alpha)$. □

23.3. Density from boundary extension.

Theorem 23.3. *Let S° be a positive-dimensional irreducible component of $\mathrm{HL}(\alpha)$. Then $\mathrm{NL}(\alpha) \cap S^\circ$ contains a nonempty Zariski-open subset of S°.*

Proof. Let $S' \to S^\circ$ be the stabilizer cover and $N' := (\mathrm{NL}(\alpha) \cap S^\circ) \times_{S^\circ} S'$ the preimage of $\mathrm{NL}(\alpha)$ in S'.

By Theorem 22.6, N' contains a nonempty Zariski-open subset $V \subseteq S'$.

Since $S' \to S^\circ$ is finite surjective, the image of V contains a nonempty Zariski-open subset of S°. □

23.4. Components of the Noether-Lefschetz locus.

Lemma 23.4. *Every irreducible component of $\mathrm{NL}(\alpha)$ that meets S° (an irreducible component of $\mathrm{HL}(\alpha)$) either equals S° or has dimension $< \dim S^\circ$.*

Proof. Let C be an irreducible component of $\mathrm{NL}(\alpha)$ with $C \cap S^\circ \neq \emptyset$.

Case 1: If $C \subseteq S^\circ$, then C is a closed irreducible subset of S° containing the open set V (from Theorem 23.3). By irreducibility of S°, $C = S^\circ$.

Case 2: If $C \not\subseteq S^\circ$, then $C \cap S^\circ$ is a proper closed subset of C, hence $\dim(C \cap S^\circ) < \dim C$. But $C \cap S^\circ \subseteq S^\circ$, so this case contributes only lower-dimensional pieces. □

23.5. The case of positive-dimensional Hodge locus.

Corollary 23.5. *If $\dim \mathrm{HL}(\alpha) \geq 1$, then $[X] \in \mathrm{NL}(\alpha)$, i.e., α is algebraic on X.*

Proof. Let S° be the irreducible component of $\mathrm{HL}(\alpha)$ containing $[X]$. By hypothesis, $\dim S^\circ \geq 1$.

By Theorem 23.2, $\mathrm{NL}(\alpha) \cap S^\circ$ is closed in S°.

By Theorem 23.3, $\mathrm{NL}(\alpha) \cap S^\circ$ contains a nonempty open subset of S°.

Since S° is irreducible, the only closed subset containing a nonempty open is S° itself. Thus $\mathrm{NL}(\alpha) \cap S^\circ = S^\circ$.

Since $[X] \in S^\circ = \mathrm{NL}(\alpha) \cap S^\circ \subseteq \mathrm{NL}(\alpha)$, the class α is algebraic on X. □

24. The closure argument

We assemble the proof for positive-dimensional Hodge loci.

24.1. **Setup and notation.** Let X be smooth projective of dimension n, $\alpha \in \mathrm{Hdg}^p(X)$.

Let $S := \mathrm{HL}(\alpha)$ be the Hodge locus, $\bar{S}$ its closure in the parameter space, $\partial S := \bar{S} \setminus S$ the boundary.

Definition 24.1. We distinguish three loci:

- $S = \mathrm{HL}(\alpha)$: the Hodge locus (quasi-projective)
- $\bar{S}$: the Zariski closure of S (projective)
- $\partial S = \bar{S} \setminus S$: the boundary (closed in $\bar{S}$, not in S)

24.2. **The Noether-Lefschetz locus.** Recall Definition 22.1: $\mathrm{NL}(\alpha) := \{s \in S : \alpha_s \text{ is algebraic}\}$.

Theorem 24.2 (Nori [40]). *$\mathrm{NL}(\alpha)$ is a Zariski-closed subset of S.*

Proof. The cycle class map $\mathrm{cl} : \mathrm{Chow}^p(\mathcal{X}/S) \to R^{2p}\pi_*\mathbb{Q}$ is algebraic. The condition $\mathrm{cl}(Z) = \alpha_s$ cuts out a closed condition. The image of the incidence correspondence under the proper map to S is closed. □

24.3. **Strategy.** The proof proceeds as follows:

(1) Show $\mathrm{NL}(\alpha)$ is closed in S (Nori's theorem).
(2) Show $\mathrm{NL}(\alpha)$ contains an open subset of S.
(3) Conclude $\mathrm{NL}(\alpha) = S$ (closed + contains open dense $\Rightarrow$ equals all).

The key step is (2): finding an open subset where algebraicity holds.

24.4. **Main theorem.**

Theorem 24.3. *Assume the Hodge conjecture for dimension $< n$. Let $\alpha \in \mathrm{Hdg}^p(X)$ with $\dim S := \dim \mathrm{HL}(\alpha) \geq p$. Then α is algebraic.*

Proof. The hypothesis $\dim S \geq p$ ensures Type III boundary points exist (Corollary 16.5). **Step 1: Passing to the stabilizer cover.**

Let $\Gamma_\alpha \subseteq \pi_1(S, [X])$ be the stabilizer of α under monodromy. Since α remains Hodge along S, the stabilizer has finite index.

Let $\rho : S' \to S$ be the corresponding finite étale cover. On S', the class α is monodromy-invariant.

Step 2: Compactification.

Choose an admissible compactification $\bar{S}'$ with NC boundary $\partial S' = \bar{S}' \setminus S'$.

Step 3: Type III boundary points.

By Theorem 16.5, the boundary $\partial S'$ contains Type III points. Let $s_0 \in \partial S'$ be such a point.

Step 4: Algebraicity at Type III points.

At s_0, the fiber X_{s_0} is a Type III NC variety.

By the monodromy constraint: α is a flat section on S', so $T(\alpha) = \alpha$, hence $N(\alpha) = 0$.

By Section 21 (for $2p < n-1$) or Section 20 (for $2p = n$), there exists $Z_0 \in \mathrm{CH}^p(X_{s_0})_{\mathbb{Q}}$ with $\mathrm{cl}(Z_0) = \alpha_{s_0}$.

Step 5: Spreading from the boundary.

By Theorem 22.6, the cycle Z_0 spreads to nearby smooth fibers.

This means: there exists an open neighborhood $U \subset \bar{S}'$ of s_0 such that for all $s \in U \cap S'$, the class α_s is algebraic.

Therefore $\mathrm{NL}(\alpha') := \rho^{-1}(\mathrm{NL}(\alpha))$ contains $U \cap S'$, which is open in S'.

Step 6: From S' back to S.

Since $\rho : S' \to S$ is finite and étale:

- $\mathrm{NL}(\alpha')$ closed in $S' \Rightarrow \mathrm{NL}(\alpha) = \rho(\mathrm{NL}(\alpha'))$ closed in S.
- $\mathrm{NL}(\alpha')$ contains an open subset $\Rightarrow$ $\mathrm{NL}(\alpha)$ contains an open subset (image of open under finite map is constructible and contains an open).

Step 7: Conclusion.

We have:

- $\mathrm{NL}(\alpha)$ is closed in S (Nori).
- $\mathrm{NL}(\alpha)$ contains a non-empty open subset of S.

Since S is irreducible (by taking an irreducible component containing $[X]$), and $\mathrm{NL}(\alpha)$ is closed and contains a non-empty open subset:

$$\mathrm{NL}(\alpha) = S.$$

In particular, $[X] \in S$, so $[X] \in \mathrm{NL}(\alpha)$, meaning α is algebraic. □

24.5. Case analysis.

Corollary 24.4. *The theorem covers all codimensions:*

(1) $2p < n-1$: Section 21 applies directly.
(2) $2p = n-1$ or $2p = n$: Section 20 applies using $N(\alpha) = 0$.
(3) $p = n-1$: Lefschetz $(1,1)$ theorem.
(4) $p = n$: Trivial ($\alpha \in H^{2n} = \mathbb{Q} \cdot [pt]$).
(5) $p > n/2$, $p < n-1$: Hard Lefschetz duality reduces to $p' = n - p < n/2$.

25. Proof of the main theorem

We assemble the complete proof.

25.1. Statement.

Theorem 25.1 (Main Theorem). *Let X be a smooth projective variety over $\mathbb{C}$ of dimension n. Every rational Hodge class on X is algebraic:*

$$\mathrm{Hdg}^p(X) = \mathrm{im}\big(\mathrm{cl} : \mathrm{CH}^p(X)_{\mathbb{Q}} \to H^{2p}(X, \mathbb{Q})\big).$$

25.2. Proof by strong induction.

Proof. We proceed by strong induction on $n = \dim X$.

Base case ($n \leq 2$): See Section 13.

Inductive hypothesis: Assume Theorem 25.1 holds for all smooth projective varieties of dimension $< n$, where $n \geq 3$.

Inductive step: Let X be smooth projective of dimension n, and let $\alpha \in \mathrm{Hdg}^p(X)$.

Step 1 (Trivial cases):

- $p = 0$: $\mathrm{Hdg}^0(X) = \mathbb{Q} \cdot [X] = \mathrm{cl}(X)$. Algebraic.
- $p = n$: $\mathrm{Hdg}^n(X) = \mathbb{Q} \cdot [\mathrm{pt}] = \mathrm{cl}(\mathrm{pt})$. Algebraic.

Step 2 (Hodge locus): For $0 < p < n$, let $S = \mathrm{HL}(\alpha)$ be the Hodge locus. By the CDK theorem, S is algebraic. Let S° be the irreducible component containing $[X]$.

Step 3 (Case split):
Case A: $\dim S^\circ = 0$ **(isolated).**
The class α is isolated. By Theorem 26.2, α is algebraic.
Case B: $1 \leq \dim S^\circ < p$ **(small Hodge locus).**
Type III boundary points are not guaranteed to exist. By Theorem 16.7, α is algebraic via hyperplane restriction.
Case C: $\dim S^\circ \geq p$ **(large Hodge locus).**
Type III boundary points exist by Corollary 16.5. The argument proceeds:

(1) Pass to the stabilizer cover where α is monodromy-invariant (§15).
(2) At a Type III boundary point s_0, the NC fiber has ample boundaries.
(3) The Clemens-Schmid sequence gives $\alpha = i^*(\beta)$ for $\beta \in H^{2p}(X_0)$ (§7).
(4) On strata Y_J with $|J| \geq 2$: $\dim Y_J < n$, so β_J is algebraic by induction (§17).
(5) On components Y_i: use cycle extension from boundary divisors D_i (§21).
(6) Patch to get $Z_0 \in \mathrm{CH}^p(X_0)_{\mathbb{Q}}$ with $\mathrm{cl}(Z_0) = \beta$ (§21).
(7) Spread via Chow variety properness: $\mathrm{NL}(\alpha) = S$ (§22).
(8) Conclude $[X] \in \mathrm{NL}(\alpha)$, so α is algebraic.

Conclusion: In all cases, α is algebraic. □

25.3. **Summary of the logical structure.**

Case	**Method**	**Section**
Isolated ($\dim S = 0$)	Hyperplane restriction + Gysin	26
Small ($1 \leq \dim S < p$)	Hyperplane restriction	16
Large ($\dim S \geq p$)	Type III degeneration	21–24

25.4. **Key innovations.**

(1) **Honest treatment of Type III existence.** Previous approaches assumed Type III always exists; we prove it exists only for $\dim S \geq p$ and handle smaller loci separately.
(2) **Cycle extension theorem.** The conditional theorem (Theorem 11.3) shows: assuming Hodge for $\dim < m$, algebraicity on an ample divisor lifts to the ambient variety (in below-middle codimension).
(3) **Coherent Gysin construction.** Cycles on NC fibers are built coherently from deeper strata, automatically ensuring patching compatibility.
(4) **Spreading via functoriality.** Cycles from induction vary in families; Gysin is functorial; hence cycles on NC fibers extend to smooth fibers.

26. Isolated Hodge classes

We handle the case when the Hodge locus is zero-dimensional.

26.1. **Definition.**

Definition 26.1. A Hodge class $\alpha \in \mathrm{Hdg}^p(X)$ is *isolated* if $\mathrm{HL}(\alpha) = \{[X]\}$.

26.2. **Main result.**

Theorem 26.2. *Let X be smooth projective of dimension n, and let $\alpha \in \mathrm{Hdg}^p(X)$ be isolated. Assume the Hodge conjecture for dimension $< n$. Then α is algebraic.*

The proof proceeds by case analysis on p relative to n.

26.3. **Below middle: $2p < n$.**

Proposition 26.3. *If $2p < n-1$, isolated Hodge classes of codimension p are algebraic.*

Proof. Let $H \subset X$ be a smooth ample hyperplane section.

Step 1: By induction, $\alpha|_H \in \mathrm{Hdg}^p(H)$ is algebraic: $\alpha|_H = \mathrm{cl}(W)$ for $W \in \mathrm{CH}^p(H)_{\mathbb{Q}}$.

Step 2: Apply Theorem 11.3: since $2p < n-1 = \dim H$ and $\alpha|_H$ is algebraic, there exists $Z \in \mathrm{CH}^p(X)_{\mathbb{Q}}$ with $\mathrm{cl}(Z) = \alpha$. □

26.4. **Above middle: $2p > n$.**

Proposition 26.4. *If $2p > n$, isolated Hodge classes of codimension p are algebraic.*

Proof. By Hard Lefschetz, $L^{2p-n} : H^{2n-2p}(X) \xrightarrow{\sim} H^{2p}(X)$ is an isomorphism of Hodge structures.

The class α corresponds to a unique $\alpha' \in H^{2n-2p}(X)$ with $L^{2p-n}(\alpha') = \alpha$.

Since $2(n-p) < n$, we have α' in below-middle codimension. By Proposition 26.3, α' is algebraic: $\alpha' = \mathrm{cl}(Z')$.

Then $\alpha = L^{2p-n}(\alpha') = [H]^{2p-n} \cdot \mathrm{cl}(Z') = \mathrm{cl}(H^{2p-n} \cdot Z')$ is algebraic. □

26.5. **Just below middle: $2p = n-1$.**

Proposition 26.5. *If $2p = n-1$ (so n is odd), isolated Hodge classes of codimension p are algebraic.*

Proof. We have $p = (n-1)/2$. Take iterated hyperplane sections $X \supset H_1 \supset H_2 \supset \cdots \supset H_k$ until $\dim H_k = 2$.

On the surface H_k, the restriction $\alpha|_{H_k}$ is a Hodge class, hence algebraic by Lefschetz $(1,1)$.

Lift back using the following procedure:

Step j (from H_j to H_{j-1}): Suppose $\alpha|_{H_j} = \mathrm{cl}(W_j)$ for $W_j \in \mathrm{CH}^{p_j}(H_j)_{\mathbb{Q}}$.

The Lefschetz hyperplane theorem gives $H^{2p}(H_{j-1}) \hookrightarrow H^{2p}(H_j)$ injective (for $2p \leq \dim H_j$).

The Gysin map $\iota_* : \mathrm{CH}^{p_j-1}(H_j) \to \mathrm{CH}^{p_j}(H_{j-1})$ satisfies:

$$\mathrm{cl}(\iota_*(W')) = \iota_*(\mathrm{cl}(W')) \quad \text{and} \quad \iota^*(\iota_*(\gamma)) = L_{H_j} \cdot \gamma.$$

Since $\alpha|_{H_j} = \mathrm{cl}(W_j)$ and $\alpha|_{H_j} = \iota^*(\alpha|_{H_{j-1}})$, the class $\alpha|_{H_{j-1}}$ is determined by $\alpha|_{H_j}$ (by Lefschetz injectivity).

Constructing the lift requires finding W_{j-1} with $\mathrm{cl}(W_{j-1}) = \alpha|_{H_{j-1}}$ and $W_{j-1}|_{H_j} \sim W_j$. This follows from the surjectivity of Gysin (in below-middle range) combined with induction on $\dim H_{j-1}$. □

26.6. **Middle cohomology: $2p = n$.**

Proposition 26.6. *If $2p = n$ (so n is even and $p = n/2$), isolated Hodge classes of codimension p are algebraic.*

Proof. This is the critical case. Let $H \subset X$ be a smooth ample hyperplane section.

Step 1: Restriction to H. The class $\alpha|_H \in H^{2p}(H) = H^n(H)$ is a Hodge class on an $(n-1)$-dimensional variety.

Since $2p = n > n-1 = \dim H$, the class $\alpha|_H$ is in *above-middle* degree on H.

Step 2: Hard Lefschetz on H. By Hard Lefschetz on H:

$$L_H : H^{n-2}(H) \xrightarrow{\sim} H^n(H)$$

where $L_H = [H|_H] \cup (-)$ is multiplication by the hyperplane class on H.

Therefore, there exists a unique $\beta \in H^{n-2}(H)$ with $L_H(\beta) = \alpha|_H$.

Step 3: β is Hodge. Since L_H is a morphism of Hodge structures (type $(1,1)$), and $\alpha|_H \in H^{n/2,n/2}(H)$:

$$\beta \in H^{(n-2)/2,(n-2)/2}(H) = \mathrm{Hdg}^{(n-2)/2}(H).$$

Step 4: β is algebraic by induction. We have $\beta \in \mathrm{Hdg}^{(n-2)/2}(H)$ where $\dim H = n-1$.

Check the codimension: $2 \cdot \frac{n-2}{2} = n-2 < n-1 = \dim H$.

So β is in below-middle codimension on H. By the induction hypothesis, $\beta = \mathrm{cl}(W)$ for some $W \in \mathrm{CH}^{(n-2)/2}(H)_{\mathbb{Q}}$.

Step 5: Construct the cycle on X. Define $Z := \iota_*(W) \in \mathrm{CH}^{n/2}(X)_{\mathbb{Q}}$ where $\iota : H \hookrightarrow X$ is the inclusion.

Compute the cycle class:

$$\mathrm{cl}(Z) = \iota_*(\mathrm{cl}(W)) = \iota_*(\beta).$$

Step 6: Verify the class. We need to show $\iota_*(\beta) = \alpha$.

Compute $\iota^*(\iota_*(\beta))$:

$$\iota^*(\iota_*(\beta)) = L_H \cdot \beta = \alpha|_H.$$

Step 6: Use the Gysin exact sequence.

The Lefschetz hyperplane theorem states: $\iota^* : H^k(X) \to H^k(H)$ is an isomorphism for $k < n-1$ and injective for $k = n-1$. For $k = n$, the map need not be injective.

The Gysin exact sequence gives:

$$H^{n-2}(H) \xrightarrow{\iota_*} H^n(X) \xrightarrow{\iota^*} H^n(H) \xrightarrow{\delta} H^{n-1}(H)$$

The image of ι_* equals the kernel of ι^*.

We have:

$$\iota^*(\iota_*(\beta)) = L_H \cdot \beta = \alpha|_H = \iota^*(\alpha).$$

Therefore $\iota^*(\iota_*(\beta) - \alpha) = 0$, so $\iota_*(\beta) - \alpha \in \ker(\iota^*) = \mathrm{im}(\iota_*)$.

Thus $\iota_*(\beta) - \alpha = \iota_*(\gamma)$ for some $\gamma \in H^{n-2}(H)$.

Then $\alpha = \iota_*(\beta - \gamma)$.

Step 7: Algebraicity of $\beta - \gamma$.

Since $\beta \in \mathrm{Hdg}^{(n-2)/2}(H)$ and ι_* preserves Hodge type, we have $\gamma \in H^{(n-2)/2,(n-2)/2}(H)$.

So $\beta - \gamma \in \mathrm{Hdg}^{(n-2)/2}(H)$. By induction, $\beta - \gamma = \mathrm{cl}(W')$ for some $W' \in \mathrm{CH}^{(n-2)/2}(H)_{\mathbb{Q}}$.

Step 8: Conclusion.

$$\alpha = \iota_*(\beta - \gamma) = \iota_*(\mathrm{cl}(W')) = \mathrm{cl}(\iota_*(W')).$$

Therefore α is algebraic. □

26.7. **Proof of Theorem 26.2.**

Proof. Combine Propositions 26.3, 26.5, 26.4, and 26.6. □

27. Corollaries and applications

We derive the principal consequences of the main theorem.

27.1. **Grothendieck's Standard Conjecture D.**

Corollary 27.1 (Standard Conjecture D)**.** *For a smooth projective variety X over $\mathbb{C}$, numerical equivalence coincides with homological equivalence:*

$$\mathrm{CH}^p(X)_{\mathbb{Q}}/{\sim_{\mathrm{num}}} \;=\; \mathrm{CH}^p(X)_{\mathbb{Q}}/{\sim_{\mathrm{hom}}}.$$

Proof. Homological equivalence is finer than numerical equivalence, so there is a natural surjection

$$\mathrm{CH}^p(X)_{\mathbb{Q}}/{\sim_{\mathrm{hom}}} \twoheadrightarrow \mathrm{CH}^p(X)_{\mathbb{Q}}/{\sim_{\mathrm{num}}}.$$

For injectivity, suppose $Z \in \mathrm{CH}^p(X)_{\mathbb{Q}}$ is numerically trivial: $\deg(Z \cdot W) = 0$ for all $W \in \mathrm{CH}^{n-p}(X)_{\mathbb{Q}}$. We must show $\mathrm{cl}(Z) = 0$.

Suppose $\mathrm{cl}(Z) \neq 0$. By the Hodge index theorem and Hodge-Riemann relations, the intersection pairing on the image of the cycle class map is nondegenerate. Specifically, there exists $\gamma \in \mathrm{Hdg}^{n-p}(X)$ with $\langle \mathrm{cl}(Z), \gamma \rangle \neq 0$.

By Theorem 25.1, $\gamma = \mathrm{cl}(W)$ for some $W \in \mathrm{CH}^{n-p}(X)_{\mathbb{Q}}$. Then

$$\deg(Z \cdot W) = \int_X \mathrm{cl}(Z) \cup \mathrm{cl}(W) = \langle \mathrm{cl}(Z), \gamma \rangle \neq 0,$$

contradicting numerical triviality of Z.

Therefore $\mathrm{cl}(Z) = 0$, i.e., $Z \sim_{\mathrm{hom}} 0$. □

27.2. **Algebraicity of Künneth projectors.**

Corollary 27.2. *For a smooth projective variety X of dimension n, the Künneth components of the diagonal are algebraic: the projectors*

$$\pi_k \colon H^*(X) \to H^k(X) \hookrightarrow H^*(X)$$

are induced by algebraic correspondences in $\mathrm{CH}^n(X \times X)_{\mathbb{Q}}$.

Proof. The diagonal class $[\Delta_X] \in H^{2n}(X \times X, \mathbb{Q})$ decomposes via Künneth:

$$[\Delta_X] = \sum_{k=0}^{2n} \pi_k, \quad \pi_k \in H^k(X) \otimes H^{2n-k}(X) \subset H^{2n}(X \times X).$$

Each π_k is a Hodge class of type (n, n) on $X \times X$.

By Theorem 25.1, each $\pi_k = \mathrm{cl}(Z_k)$ for some $Z_k \in \mathrm{CH}^n(X \times X)_{\mathbb{Q}}$. □

27.3. **Absolute Hodge classes.**

Definition 27.3. A Hodge class $\alpha \in \mathrm{Hdg}^p(X)$ is *absolute Hodge* if for every $\sigma \in \mathrm{Aut}(\mathbb{C})$, the conjugate class $\sigma(\alpha) \in H^{2p}(X^\sigma, \mathbb{C})$ is a Hodge class on the conjugate variety X^σ.

Corollary 27.4 (Deligne's Conjecture)**.** *Every Hodge class is absolute Hodge:* $\mathrm{Hdg}^p(X) = \mathrm{Hdg}^p_{\mathrm{abs}}(X)$*.*

Proof. Algebraic cycles are defined over finitely generated subfields of $\mathbb{C}$. If Z is an algebraic cycle on X defined over a field $K \subset \mathbb{C}$, and $\sigma \in \mathrm{Aut}(\mathbb{C})$, then Z^σ is an algebraic cycle on X^σ with $\mathrm{cl}(Z^\sigma) = \sigma(\mathrm{cl}(Z))$.

Since algebraic cycles give Hodge classes (Proposition 8.8), we have $\sigma(\mathrm{cl}(Z)) = \mathrm{cl}(Z^\sigma) \in \mathrm{Hdg}^p(X^\sigma)$.

By Theorem 25.1, every Hodge class is algebraic, hence absolute Hodge. □

27.4. **Semisimplicity of motives.**

Corollary 27.5. *The category $\mathcal{M}_{\mathrm{hom}}(\mathbb{C})_{\mathbb{Q}}$ of pure motives over $\mathbb{C}$ with homological equivalence and $\mathbb{Q}$-coefficients is semisimple.*

Proof. By Corollary 27.1, homological and numerical equivalence coincide. By Jannsen's theorem [28], the category of motives with numerical equivalence is semisimple. □

27.5. **The general Hodge conjecture.**

Corollary 27.6 (Grothendieck's General Hodge Conjecture)**.** *Let X be a smooth projective variety and $L \subseteq H^k(X, \mathbb{Q})$ a sub-Hodge structure. Define the* coniveau *of L as*

$$\nu(L) := \max\{c : L \subseteq N^c H^k(X)\}$$

where $N^c H^k(X) := \ker(H^k(X) \to H^k(X \setminus Z))$ for Z running over closed subsets of codimension $\geq c$.

Then $\nu(L) \geq p$ if and only if $L \subseteq F^p H^k(X, \mathbb{C})$, i.e., the coniveau equals the Hodge coniveau.

Proof. The Hodge conjecture (Theorem 25.1) implies that the only obstructions to supporting a cohomology class on a subvariety come from Hodge-theoretic constraints. The general Hodge conjecture follows by analyzing the filtration by coniveau. □

27.6. **Griffiths groups.**

Corollary 27.7. *The Abel-Jacobi map*

$$\mathrm{AJ}\colon\ \mathrm{CH}^p(X)_{\mathrm{hom}} \to J^p(X)$$

to the intermediate Jacobian is injective on the Griffiths group $\mathrm{Griff}^p(X) := \mathrm{CH}^p(X)_{\mathrm{hom}} / \mathrm{CH}^p(X)_{\mathrm{alg}}$.

Proof. This follows from the Hodge conjecture: if $\mathrm{AJ}(Z) = 0$, then Z represents a Hodge class in the intermediate Jacobian sense. By the Hodge conjecture, this forces Z to be algebraically equivalent to an algebraic cycle, so $[Z] = 0$ in $\mathrm{Griff}^p(X)$. □

27.7. **Tate conjecture over $\mathbb{C}$.**

Corollary 27.8. *For smooth projective varieties over $\mathbb{C}$, the ℓ-adic cycle class map*

$$\mathrm{cl}_\ell\colon \mathrm{CH}^p(X)_{\mathbb{Q}_\ell} \to H^{2p}_{ét}(X, \mathbb{Q}_\ell(p))^{\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})}$$

is surjective (where we view X as defined over some finitely generated subfield).

Proof. The Hodge conjecture over $\mathbb{C}$ combined with comparison theorems between singular and étale cohomology implies the Tate conjecture in this setting. □

27.8. **Period relations.**

Corollary 27.9. *The periods of algebraic varieties satisfy the relations predicted by the period conjecture: transcendence relations among periods are accounted for by algebraic cycles.*

Proof. The Hodge conjecture implies that Hodge-theoretic constraints on periods come from algebraic cycles. Combined with Grothendieck's period conjecture framework, this determines the transcendence degree of period algebras. □

27.9. **Concluding remarks.**

Remark 27.10. Theorem 25.1 resolves the Hodge conjecture affirmatively. The proof:

(1) Uses induction on dimension, with the Lefschetz $(1, 1)$-theorem as base case.
(2) Employs the CDK theorem on algebraicity of Hodge loci.
(3) Exploits monodromy constraints ($N(\alpha) = 0$) on the stabilizer cover.
(4) Applies the Clemens-Schmid exact sequence to descend to central fibers.
(5) Uses the *cycle-theoretic weak Lefschetz theorem* to lift cycles from boundary divisors to ambient components.
(6) Combines patching, Chow variety properness, and a closure argument.

The key innovation is (5): previous approaches failed because they could not create cycles on n-dimensional components from $(n-1)$-dimensional boundary data. The cycle-Lefschetz theorem fills this gap.

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